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CHARACTERIZATION OF PLASMONIC DISORDERED METASURFACES FOR BIOSENSING APPLICATIONS UNDER REALISTIC CONDITIONS

PROTOCOLO DE INVESTIGACIÓN

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Background and Objectives of the Project

Background

Optical metasurfaces are bidimensional ensembles of metallic or dielectric nanostructures tailored to exhibit optical responses not found in naturally-occurring materials when illuminated at selected wavelengths [1, 2]. Their optical properties are determined by the physical characteristics of their constituent nanostructures, such as size, shape, orientation, and spatial distribution within the ensemble [1, 3]. The tunability achieved through these design parameters enables precise spatial control of light, positioning metasurfaces as versatile tools for applications in spectroscopy [1], color generation [2], optical communications [4], and optical sensing [3, 5, 6].

In recent years, the use of metasurfaces in biomedical and diagnostic technologies has gain interest due to their potential for highly sensitive, real-time, and label-free detection [5, 7]. Particularly, plasmonic metasurfaces have proven to be effective as refractive index sensors for biomolecular detection [1, 7–13]. For instance, Kabashin et al. [8] designed a biosensing-aimed metasurface consisting of an ordered array of cylindrical gold (Au) nanoparticles (NPs) that detects refractive index changes via spectral shifts in reflectivity minima [Fig. 1a)]. More recently, Qiu et al. [13] employed a disordered metasurfaces of spherical AuNPs to detect SARS-CoV-2

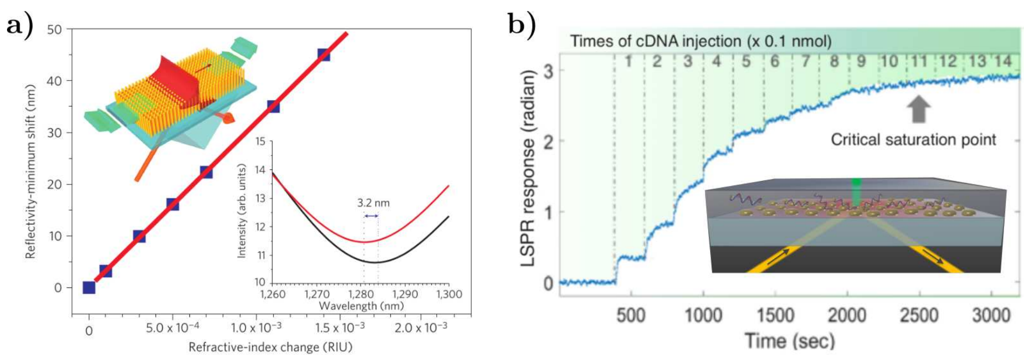

Fig. 1: Examples of biosensing-aimed plasmonic metasurfaces. **a)** Theoretical reflectivity minimum shift of an ordered metasurface of Au nanorods (scheme in the inset) as a function of the refractive index of the media embedding the nanorods; adapted from Ref. [8]. **b)** Real time measurements of the LSPR phase response obtained by illuminating a disordered metasurface of Au nanospheres (scheme in the inset) at different thiol-cDNA concentrations for SARS-CoV-2 detection; adapted from Ref. [13].

DNA sequences by monitoring phase shifts in the reflected light at wavelengths exciting the the Localized Surface Plasmon Resonance (LSPR) [Fig. 1b)]. The two theoretical-experimental works presented in Fig. 1 are not only examples of different kinds of metasurfaces, but also of different sensing mechanisms: one based on reflectivity [8] and the other on ellipsometric [13] measurements. The design of a metasurface and the choice of its sensing mechanism considers not only fabrication feasibility but also the targeted optical response, which can be predicted through different theoretical approaches.

In terms of fabrication, the disordered metasurfaces of spheroidal NPs have gained attention due to their ease of synthesis through scalable techniques such as thermal dewetting [9, 14] and laser ablation [15]. In particular, the dewetting is a fabrication method, which consists in thermally annihilating metallic thin films to synthesize spheroidal NPs [14], which exhibit broad size distributions and random spatial arrangements [14, 15]. Fig. 2a) presents Scanning Electron Microscopy (SEM) images and size distributions of synthesized AuNPs ensembles fabricated by dewetting Au films with a thickness of 5 nm (in red) and a 60 nm (in blue). As exemplified with Fig. 2a), the dewetting is a method that reduces production cost and time [9], specially when compared to lithography techniques for ordered metasurfaces [16]. Techniques like dewetting enables large-area fabrication, making disordered metasurfaces attractive for practical biosensing applications [9, 10, 13, 14].

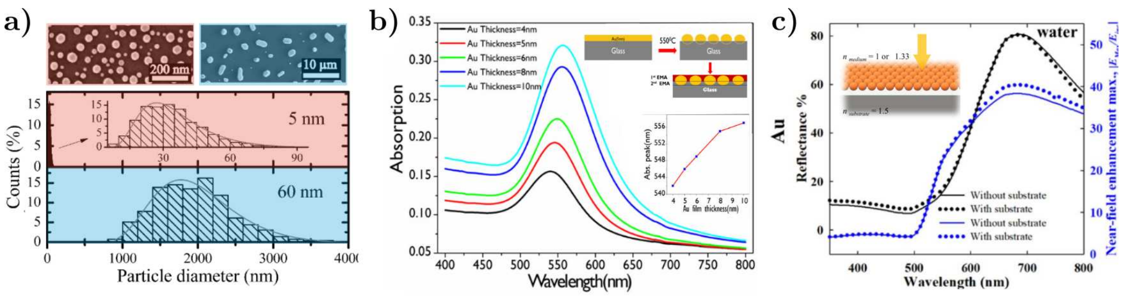

Fig. 2: a) Size distribution and SEM images of AuNPs synthesized by dewetting a 5 nm thick gold film on a silicon substrate (red) and a 60 nm thick gold film (blue); adapted from Ref. [14]. **a)** Experimental absorption spectra of AuNPs ensembled fabricated through the dewetting of gold films with varying thickness. The maxima of each curve is presented in the inset plot as a function of the film thickness and along a depiction of the theoretical model to reproduce the experimental data consisting in two layers of an EMT; adapted from Ref. [10]. **c)** Reflectance (black) and near-field enhancement maxima spectra at normal incidence of a AuNPs metasurface embedded in water in the absence (continuous lines) and in the presence (disk markers) of a glass substrate. Calculations performed in COMSOL Multiphysics$^{\text{TM}}$ considering AuNPs with a radius of 10 nm in a closed packed hexagonal array with an interparticle distance of 1nm; adapted from Ref. [17].

The theoretical response of metasurfaces aimed to biosensign have been determined with both numerical and analytical methods. Numerical simulations based on the Finite Element Method (FEM) [11, 17] or the Finite-Difference Time-Domain (FDTD) method [18] have been widely used for periodic structures. However, these methods are computationally demanding when applied to disordered systems [18, 19]. To address disordered meatasurfaces analytically, two primarily categories arises: Effective Medium Theories (EMTs) and Multiple Scattering Theories (MSTs). Some EMTs used to model metasurfaces conformed of spheroidal particles are the Maxwell Garnett Model [20] and the Bruggeman Formula [21], whose derivation considers colloidal 3D systems under the quasistatic approximation [22]. While phenomenological modifications have been implemented to suit these EMTs to metasurfaces [8], the MSTs provide a more rigorous and realistic framework by modeling the metasurface as a discrete collection of interacting particles up to different orders of approximation [23]. The Coherent Scattering Model (CSM) [20] and the Thin Film Theory (TFT) [21] are two MSTs capable of accounting for realistic conditions derived from the fabrication process, such as size distributions and substrate interactions [21, 24].

As stated above, metasurfaces fabricated by dewetting can exhibit a borad size distribution of the synthesizes NPs and a random spatial arrange of them [14, 15]. Although both the CSM and TFT are suitable for describing these systems, certain fabrication-related effects —such as partial embedding of NPs into the substrate— are not yet fully considered by current MSTs. This partial embedding, observed for specific combinations of film thickness and annealing temperature [10, 25], can significantly alter the optical properties, as evidenced by spectral shifts in experimental extinction spectra [Fig. 2b)]. Despite attempts to account for substrate effects using phenomenological EMT-based fittings, as illustrated in the inset of Fig. 2b), a comprehensive theoretical understanding of this kind of substrate interactions in disordered metasurfaces remains lacking.

While the influence of the substrate on single NPs has been rigorously studied [26–28] its impact on ensembles becomes a less explored area. Numerical methods have been proved useful for these studies when applied to ordered systems. For instance, FEM simulations —and their respective experimental corroboration— were conducted by Borah et al. [17] for AuNPs in a hexagonal array, reporting the effect on the optical response of the ensemble due to its density, illumination scheme and even the effect of the substrate when illuminated. In particular, Fig. 2c) depicts the substrate effect by sowing the reflectivity and near-field enhancement maxima spectra of a closed packed hexagonal AuNPs array illuminated at normal incidence when a substrate is considered (disk markers) or not (continuous lines). However, analogous studies for disordered arrays are scarce, leaving open questions about how substrate interactions, including partial embedding, influence the collective optical response and how these effects impact the design and performance of metasurfaces for sensing applications.

Objectives of the Project

The project presented in this work arises from the collaboration between the Nanoplasmonics Group —focused on theoretical research and led by Prof. Alejandro Reyes Coronado— at Facultad de Ciencias, UNAM and the Biophotonics Group —focused on experimental research and led by Prof. Rubén Ramos García and Prof. Svetlana Mansurova— at the Instituto Nacional de Astrofísica, Óptica y Electronica (INAOE). The goal of this collaboration is to design, characterize —theoretically and experimentally— a disordered metasurface of spheroidal AuNPs aimed for biosensing applications, considering the most realistic synthesis constrains provided by the Biophotonics Group.

The **general objective** of this work is to theoretically identify and quantify the contributions of substrate effects and multiple scattering to the optical response of disordered metasurfaces validating against experimental data. To achieve this goal, the following **specific objectives** are explored in this work:

- **Implementation of Analytical and Numerical Techniques**
The first objective consists in the development of numerical code and the use of specialized software to model the response of disordered metasurfaces. The derivation of the CSM and the TFT are presented in Sections 1.1 and 1.2, respectively, to provide a more comprehensive interpretation of their predictions. Experimental reflectivity and extinction measurements, obtained using the setups described in Section 1.3, were used to contrast these predictions. Furthermore, the Barzilai–Borwein gradient descent algorithm [29] was implemented for parameter fitting and FEM simulations in COMSOL Multiphysics$^{\text{TM}}$ ver. 6.1 were performed.

- **Quantification of Multiple Scattering and Substrate Effects**
The second objective is the application of the developed CSM code and of the numerical methods to quantify the effects of the multiple scattering and of the substrate in the optical response of an ensemble of AuNPs, analogous to the work of Borah et al. [17]. In Section 2.1 the optical response of AuNPs metasurfaces synthesized and optically characterized by the Biophotonics Group were analyzed. Results suggest that a partial embedding of the AuNPs is the primary factor behind the observed spectral shifts, as confirmed through CSM calculations, parameter fitting, and FEM simulations of single particles and dimers. Additionally, first measurements for partially embeded metasurfaces —conducted during a short research fellowship at INAOE— are presented in Section 2.2 in support of these findings.

- **Design proposal and spectral analysis of biosensing-aimed metasurfaces**
The last objective of this work is to explore the spectral response of two metasurfaces with distinct morphologies, using both the CSM and TFT, accordingly. In Section 3.1 the Topological Darkness phenomenon [30] is introduced as the sensing mechanism in disordered metasurfaces as refractive index sensor and a comparison between a reflectivity-based and a pahse-based sensing scheme was performed. Notably, a spectral blueshift in response to increasing refractive index —a phenomenon not previously reported— is predicted. A preliminary experimental validation of this result is provided in Section 3.2.

Chapter 1

Theoretical Frameworks and Measurement Techniques

The study of wave propagation in complex media has long been a central topic in optics, acoustics, and condensed matter physics. In particular, understanding the optical response of ensembles of nanoparticles (NPs) has garnered significant attention due to its relevance in fields such as nanophotonics, plasmonics, and metamaterials [7, 8, 13]. The Multiple Scattering Theories (MSTs) are theoretical frameworks employed to model these systems, which explicitly account for the interactions between individual scatterers and provide an accurate description of wave propagation in structured and disordered media [20, 23, 31]. Thus, the MST can be understood as approximate solutions to Maxwell's equations to the total scattered field for ensembles of multiple particles [23]. For instance, the Foldy-Lax equations, offer a rigorous way to model the self-consistent field scattered by a collection of particles [32]. Another alternative to solve the Maxwell's equation approximately is the formalism of excess fields, which is suited specifically to ensembles of particles located in a semiinfinite region confined in one direction [21].

In the following Sections, two MSTs and experimental setups are presented. In Section 1.1 the Coherent Scattering Model (CSM) —based on the Foldy-Lax approach— is derived, while in Section 1.2 the Thin Film Theory is presented —founded on the excess field formalism—, since both of these MSTs are suited to model the optical response of metasurfaces conformed by ensembles of spherical nanoparticles, whose centers are randomly located but contained within a plane parallel to a substrate supporting each element of the ensemble. Lastly, in Section 1.3 two experimental setups are presented, which enables the optical characterization of such metasurfaces: one for extinction and one for reflectivity spectra measurements.

1.1 Coherent Scattering Model

Multiple Scattering Theories (MSTs) provide approximate solutions to Maxwell's equations when considering the problem of light scattered by an ensemble of particles illuminated by a monochromatic plane wave [20, 23]. In particular, the Coherent Scattering Model (CSM) is a MST that provides analytical expressions for the amplitude coefficients of reflection and transmission of light scattered in the coherent direction for a disordered ensemble of spherical particles with coplanar centers [20, 33, 34].

Within the MST framework, the optical response of a disordered array of arbitrary particles, illuminated by an incident monochromatic plane wave $\mathbf{E}^{\text{inc}}(\mathbf{r},\omega)$ with angular frequency ω and propagation direction $\hat{\mathbf{k}}^{\text{inc}}$, is determined by the excitation field $\mathbf{E}^{\text{exc}}(\mathbf{r},\omega)$ acting on the k-th particle in a collection of N particles. This field is equal to the sum of the incident electric field and the induced electric field $\mathbf{E}_\ell^{\text{ind}}(\mathbf{r},\omega)$ —which includes both the field inside the particle and the scattered field outside of it— due to the other elements of the ensemble but not itself [33, 34], that is

$$\mathbf{E}_k^{\text{exc}}(\mathbf{r}) = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell\neq k}^{N} \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}), \tag{1.1}$$

where the harmonic dependence is implied in the used notation. Since all particles are excited by the same interaction, the electric field induced in the ℓ-th particle is given by [33, 34]

$$\mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) = \int \mathrm{d}^3r' \mathbb{G}(\mathbf{r},\mathbf{r}') \int \mathrm{d}^3r'' \mathbb{T}(\mathbf{r}'-\mathbf{r}_\ell, \mathbf{r}''-\mathbf{r}_\ell)\mathbf{E}_\ell^{\text{exc}}(\mathbf{r}''), \tag{1.2}$$

where $\mathbf{r}_\ell$ is the position of the center of the ℓ-th particle, $\mathbb{G}(\mathbf{r},\mathbf{r}')$ is the dyadic Green function —solution to the vectorial Helmholtz equation with the product of the unitary dyadic and a Dirac delta centered at $\mathbf{r}'$ as a source [35]— and $\mathbb{T}$ is the transition operator, or T-matrix, that linearly relates the scattered electric field of a particle to the electric field that excites it [35]. The integrals in Eq. (1.2) are performed over a volume V, described by the positions $\mathbf{r}'$ and $\mathbf{r}''$, such that the center $\mathbf{r}_\ell$ of the ℓ-th particle is located in V [33].

The total electric field $\mathbf{E}(\mathbf{r})$ is equal to the sum of $\mathbf{E}^{\text{inc}}(\mathbf{r})$ and the N contributions of the induced electric field $\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})$ for each particle, which is determined by solving the system of N equations given by Ecs. (1.1) and (1.2). For an ensemble of randomly located particles, the configurational average of the total electric field $\langle\mathbf{E}(\mathbf{r})\rangle$ is calculated by considering the probability of occurrence of each combination in which the centers $\mathbf{r}_\ell$ of the elements can be located [33, 34]:

$$\langle\mathbf{E}(\mathbf{r})\rangle = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell=1}^{N}\langle\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})\rangle = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell=1}^{N}\left(\prod_{k=1}^{N}\int \mathrm{d}^3r_k \rho(\mathbf{R})\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})\right), \tag{1.3}$$

with $\rho(\mathbf{R})$ is the probability density that the ensemble is in a specific spatial configuration given by $\mathbf{R} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$. Configurationally averaging Eq. (1.2) gives the average contribution of the induced electric field by the ℓ-th particle, which can be written as [33, 34]

$$\langle\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})\rangle = \int \mathrm{d}^3r' \mathbb{G}(\mathbf{r},\mathbf{r}') \int \mathrm{d}^3r'' \int \mathrm{d}^3r_\ell \rho(\mathbf{r}_\ell) \mathbb{T}(\mathbf{r}'-\mathbf{r}_\ell, \mathbf{r}''-\mathbf{r}_\ell)\langle\mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'',\mathbf{R})\rangle_\ell, \tag{1.4a}$$

$$\langle\mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'',\mathbf{R})\rangle_\ell = \prod_{\substack{k=1\\k\neq\ell}}^{N}\int \mathrm{d}^3r_k \rho(\mathbf{R}|\mathbf{r}_\ell)\mathbf{E}_\ell^{\text{exc}}(\mathbf{r}''), \tag{1.4b}$$

where $\rho(\mathbf{r}_\ell)$ is the probability density of finding the center of the ℓ-th particle in V, and Eq. (1.4b) represents the configurational average of the excitation electric field $\mathbf{E}_\ell^{\text{exc}}$ for the ℓ-th particle, considering that its position in V is fixed, constriction described by the conditional probability density $\rho(\mathbf{R}|\mathbf{r}_\ell)$. This expression can be further analyzed by substituting Eq. (1.1) into Eq. (1.4b) and following an analogous procedure to that of Eqs. (1.4) with the indices exchanges $k \to \ell$ and $\ell \to m$, yielding

$$\begin{aligned}\langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R})\rangle_\ell =& \mathbf{E}^{\text{inc}}(\mathbf{r}'') + \sum_{\substack{m=1\\ m\neq\ell}}^{N} \int \mathrm{d}^3r' \mathbb{G}(\mathbf{r}', \mathbf{r}'')\times \\ & \int \mathrm{d}^3 r''' \int \mathrm{d}^3 r_m \rho(\mathbf{r}_m) \mathbb{T}(\mathbf{r}' - \mathbf{r}_m, \mathbf{r}''' - \mathbf{r}_m)\langle \mathbf{E}_m^{\text{exc}}(\mathbf{r}'', \mathbf{R})\rangle_{\ell,m}, \end{aligned} \tag{1.5a}$$

$$\langle \mathbf{E}_m^{\text{exc}}(\mathbf{r}'', \mathbf{R})\rangle_{\ell,m} = \prod_{\substack{n=1\\ n\neq\ell,m}}^{N} \int \mathrm{d}^3 r_n \rho(\mathbf{R}|\mathbf{r}_\ell, \mathbf{r}_m)\mathbf{E}_n^{\text{exc}}(\mathbf{r}''), \tag{1.5b}$$

where Eq. (1.5b) represents the configurational average of the excitation field $\mathbf{E}_m^{\text{exc}}(\mathbf{r}'')$ acting on the m-th particle, considering the conditional probability density $\rho(\mathbf{R}|\mathbf{r}_\ell, \mathbf{r}_m)$ that the system is in a configuration $\mathbf{R}$ with the positions $\mathbf{r}_\ell$ and $\mathbf{r}_m$ fixed in space [33, 34]. By continuing this process for all N spheres in the ensemble, a set of N equations known as the Foldy-Lax hierarchy [23] can be constructed. Solving this hierarchy leads to the solution of the averaged electric field at a point $\mathbf{r}$, taking multiple scattering effects into account. The hierarchical equations can be truncated at different orders, reducing them to invertible systems of equations by applying approximations to how multiple scattering excites the elements of the ensemble [23].

To truncate the Foldy-Lax hierarchy and determine the total averaged electric field, the CSM imposes two approximations at different orders. First, the Single Scattering Approximation (SSA) is considered, reducing the Foldy-Lax hierarchy to a single equation to be solved by neglecting multiple scattering effects [33]. Specifically, in the SSA, $\langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R})\rangle_\ell$ is forced to equal the incident electric field $\mathbf{E}^{\text{inc}}(\mathbf{r})$ in Eq. (1.4b) [23, 34]. The second approximation in the CSM is the Quasi-Crystalline Approximation (QSA), where the Foldy-Lax hierarchy is rewritten as a system of two equations by assuming that the configurational average for one and two fixed particles are approximations of each other, meaning that Eqs. (1.4b) and (1.5b) are set equal [23, 33]. While the SSA is suitable for dilute disordered arrays, as it disregards multiple scattering, the QSA adapts to denser ensembles —up to a coverage fraction of 60% [23]— because, under this condition, the average with two fixed particles does not significantly differ from the average with one fixed particle due to the presence of the remaining particles [33].

Once the approximation orders have been determined to truncate the Foldy-Lax hierarchy, the CSM introduces a series of assumptions about the particle ensemble to solve the resulting system of equations for each approximation. In particular, the CSM considers that the particle ensemble consists of $N \gg 1$ identical spherical particles of radius a —embedded in a dielectric medium known as the matrix— whose centers $\mathbf{r}_\ell$ are located within the integration volume V, which consists of an infinite film of thickness d centered at $z = 0$, with a uniform probability density $\rho(\mathbf{r}_\ell) = 1/V$, and assumes that the volume V_ℓ of each sphere is negligible compared to V [33]; a schematic of the system is shown in Fig. 1.1. Under these assumptions, the electric field induced by the ensemble particles in the SSA can be computed by substituting in Eq. (1.4) the dyadic Green function $\mathbb{G}$ with its plane wave representation and performing the configurational average, along with the limit $d \to 0$ to consider the case of a bidimensional ensemble [33]. This

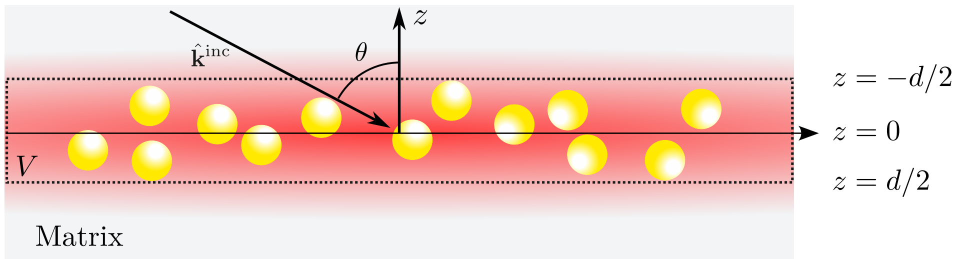

Fig. 1.1: Diagram of the particle ensemble considered in the CSM, consisting of a collection of N identical spherical particles of radius a whose centers are randomly located within the integration volume V: an infinite film centered at the plane $z = 0$ and of thickness d, which tends to zero to reproduce the case of a two-dimensional array. The system, embedded in a dielectric matrix (gray background), is illuminated by an incident monochromatic plane wave with traveling in the $\hat{\mathbf{k}}^{\rm inc}$ direction into the film at an angle θ. The red region represents the induced electric field due to the spherical particles.

procedure results in

$$\langle \mathbf{E}^{\rm ind}_{\rm SSA}(\mathbf{r})\rangle = \begin{cases} -\alpha S_j(\pi - 2\theta)\exp(i\mathbf{k}^{\rm r}_{\rm coh}\cdot\mathbf{r})\mathbf{E}^{\rm inc}(\mathbf{r}), & z > 0, \\ -\alpha S(0)\exp\left(i\mathbf{k}^{\rm t}_{\rm coh}\cdot\mathbf{r}\right)\mathbf{E}^{\rm inc}(\mathbf{r}), & z < 0, \end{cases} \quad \text{with} \quad \alpha = \frac{2\Theta}{x^2\cos\theta}, \tag{1.6}$$

where θ is the incidence angle at which the incident plane wave illuminates the two-dimensional array, $S_j(\vartheta)$ are the elements of the Mie scattering matrix [36] —which depend on the refractive index of the matrix, the sphere, and its size— evaluated at ϑ, with $j = 1$ for s-polarization and $j = 2$ for p-polarization. The term Θ corresponds to the surface coverage of the ensemble on the plane $z = 0$, and $x = k^{\rm ind}a$ is the size parameter with $k^{\rm ind}$ being the magnitude of the incident field wave vector propagating in the matrix [33, 34]. The elements $S_j(\vartheta)$ are evaluated at $\vartheta = \pi - 2\theta$ and $\vartheta = 0$ as they correspond to the coherent scattering directions given by the law of reflection and Snell's law, respectively. From Eq. (1.6), it is shown that under the SSA, the bidimensional ensemble of spherical particles scatters light primarily in coherent directions, as the diffuse components interfere destructively [33]. Consequently, reflection $r^{(j)}_{\rm SSA}$ and transmission $t^{(j)}_{\rm SSA}$ amplitude coefficients can be defined for the two-dimensional array similarly to the Fresnel coefficients, given by

$$r^{(j)}_{\rm SSA}(\theta) = -\alpha S_j(\pi - 2\theta), \qquad \text{and} \qquad t^{(j)}_{\rm SSA}(\theta) = 1 - \alpha S(0). \tag{1.7}$$

In the case of the QSA, the system of equations to be solved consists of Eqs. (1.4) and (1.5), considering that Eqs. (1.4b) and (1.5b) are equal. To solve this system of equations, the ensemble is assumed to have the same characteristics as in the SSA case and additionally that the surroundings of each sphere are identical on average [34]. Therefore, the system of equations can be solved by proposing the *Ansatz*

$$\langle \mathbf{E}^{exc}_{\ell}(\mathbf{r})\rangle_{\ell} = \mathbf{E}^{\rm exc}_{\rm r}(\mathbf{r})\ \exp(i\mathbf{k}^{\rm r}_{\rm coh}\cdot\mathbf{r}) + \mathbf{E}^{\rm exc}_{\rm t}(\mathbf{r})\ \exp\left(i\mathbf{k}^{\rm t}_{\rm coh}\cdot\mathbf{r}\right), \tag{1.8}$$

that is, in the CSM it is assumed that $\langle \mathbf{E}^{exc}_{\ell}(\mathbf{r})\rangle_{\ell}$ consists of a contribution from an electric field propagating in the coherent reflection direction, denoted as $\mathbf{E}^{\rm exc}_{\rm r}$, and another in the transmission direction, denoted by $\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}$ [34]. These coherent contributions behave according to the SSA [Eq. (1.6)], so each of them will have a reflected and transmitted component. The sum of these contributions determines the induced electric field under the QSA. Solving the system of equations and substituting in Eq. (1.9), the amplitude coefficients of reflection $r_{\mathrm{CSM}}^{(j)}$ and transmission $t_{\mathrm{CSM}}^{(j)}$ can be defined, considering multiple scattering [33, 34]. These coherent contributions behave according to the SSA [Eq. (1.6)], so each of them will have a reflected and a transmitted component. The sum of these contributions determines the induced electric field under the QSA, whose expression is

$$\langle \mathbf{E}_{\mathrm{QSA}}^{\mathrm{ind}}(\mathbf{r})\rangle = \begin{cases} -\alpha\Big(S_j(0)\|\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r})\| + S(\pi-2\theta)\|\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})\|\Big)\exp(i\mathbf{k}_{\mathrm{coh}}^{\mathrm{r}}\cdot\mathbf{r})\hat{\mathbf{E}}^{\mathrm{inc}}, & z>0, \\ -\alpha\Big(S_j(\pi-2\theta)\|\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r})\| + S(0)\|\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})\|\Big)\exp\Big(i\mathbf{k}_{\mathrm{coh}}^{\mathrm{t}}\cdot\mathbf{r}\Big)\hat{\mathbf{E}}^{\mathrm{inc}}, & z<0, \end{cases} \tag{1.9}$$

where $\hat{\mathbf{E}}^{\mathrm{inc}}$ indicates the polarization of the incident electric field. Since in the QSA the expressions of Eq. (1.5) are equal for the electric field exciting a particle, then the contributions of the induced field in the coherent direction satisfy [34]

$$\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r}) = -\frac{\alpha}{2}\Big(S_j(0)\|\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r})\| + S(\pi-2\theta)\|\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})\|\Big)\hat{\mathbf{E}}^{\mathrm{inc}}, \tag{1.10a}$$

$$\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r}) = \mathbf{E}^{\mathrm{inc}}(\mathbf{r}) - \frac{\alpha}{2}\Big(S_j(\pi-2\theta)\|\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}\| + S(0)\|\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})\|\Big)\hat{\mathbf{E}}^{\mathrm{inc}}. \tag{1.10b}$$

Solving the system of equations for $\mathbf{E}_{\mathrm{r}}^{\mathrm{exc}}(\mathbf{r})$ and $\mathbf{E}_{\mathrm{t}}^{\mathrm{exc}}(\mathbf{r})$ shown in Eq. (1.10), and substituting them in Eq. (1.9), expressions for the reflection $r_{\mathrm{CSM}}^{(j)}$ and transmission $t_{\mathrm{CSM}}^{(j)}$ amplitude coefficients are defined, considering multiple scattering. These expressions are given by [33, 34]

$$r_{\mathrm{CSM}}^{(j)}(\theta) = \frac{-\alpha S_j(\pi-2\theta)}{1+\alpha S(0)+\frac{1}{4}\alpha^2\left[S^2(0)-S_j^2(\pi-2\theta)\right]}, \tag{1.11a}$$

$$t_{\mathrm{CSM}}^{(j)}(\theta) = \frac{1-\frac{1}{4}\alpha^2\left[S^2(0)-S_j^2(\pi-2\theta)\right]}{1+\alpha S(0)+\frac{1}{4}\alpha^2\left[S^2(0)-S_j^2(\pi-2\theta)\right]}, \tag{1.11b}$$

where $j=1$ for s-polarization and $j=2$ for p-polarization. The assumption that spherical particles are identical, imposed in both the SSA and QSA, can be relaxed if the radius a of the particles follows a size distribution density $\rho(a)$ and the field exciting each sphere is identical for all, except for a phase difference of $\pm 2ik^{\mathrm{inc}}a$ caused by size variation [24]. In this case, the average over the particle radius a is computed in Eq. (1.10), incorporating the phase difference by multiplying the right-hand side of Eq. (1.10) by $\exp(\pm 2ik_z^{\mathrm{inc}}a)$ —with $k_z^{\mathrm{ind}} = k^{\mathrm{ind}}\cos\theta$—, where the positive sign corresponds to the reflected contribution and the negative to the transmitted one. Following the analogous procedure described earlier, the reflection and transmission amplitude coefficients considering size polydispersity in the bidimensional ensemble are given by [24].

$$r_{\mathrm{CSM}}^{(j)}(\theta) = \frac{-\beta_+^{(j)}(\theta)}{1+\beta_F+\frac{1}{4}\left[\beta_F^2-\beta_+^{(j)}(\theta)\beta_-^{(j)}(\theta)\right]}, \tag{1.12a}$$

$$t_{\mathrm{CSM}}^{(j)}(\theta) = \frac{1 - \frac{1}{4}\left[\beta_F^2 - \beta_+^{(j)}(\theta)\beta_-^{(j)}(\theta)\right]}{1 + \beta_F + \frac{1}{4}\left[\beta_F^2 - \beta_+^{(j)}(\theta)\beta_-^{(j)}(\theta)\right]}, \tag{1.12b}$$

where

$$\beta_F(\theta) = \eta \int_0^\infty \rho(a) S(0)\,\mathrm{d}a \quad \text{and} \quad \beta_\pm^{(j)}(\theta) = \eta \int_0^\infty \rho(a) S_j(\pi - 2\theta) \exp\left(\pm 2ik_z^{\mathrm{inc}} a\right) \mathrm{d}a\,, \tag{1.13}$$

with $\eta = 2\Theta/(x_0^2 \cos\theta)$ and where $x_0 = k^{\mathrm{ind}} a_0$ is the size parameter of a sphere with radius a_0, which is the most probable value according to the size distribution density $\rho(a)$.

The expressions in Eq. (1.12) allow the modeling of a disordered metasurface of spherical NPs under conditions close to experimental ones by introducing the presence of a substrate. To achieve this, multiple reflections between the substrate-matrix interface and the matrix-metasurface interface are considered, assuming that the latter responds according to Eq. (1.12) when separated from the substrate by a distance $\Delta \to 0$ [24]. In particular, to compare the theoretical response with reflectance measurements in an internal incidence scheme, the amplitude reflection coefficient of the entire system (metasurface with substrate) is given by [24]

$$r_{\mathrm{CSM}}^{(j)}(\theta_i) = \frac{r_{\mathrm{sm}}^{(j)}(\theta_i) + r_{\mathrm{CSM}}^{(j)}(\theta_t)}{1 + r_{\mathrm{sm}}^{(j)}(\theta_i) r_{\mathrm{CSM}}^{(j)}(\theta_t)}, \tag{1.14}$$

where r_{sm} is the Fresnel reflection coefficient between the substrate and the matrix for s (p) polarization if $j = 1$ ($j = 2$), and where θ_i is the angle of incidence of the plane wave illuminating the interface between the substrate and the matrix, and θ_t is the transmission angle upon crossing this interface and thus the angle at which it illuminates the metasurface.

1.2 Thin Film Theory

The Thin Film Theory (TFT) is a MST-based framework suited to describe the electromagnetic properties, in the far-field regime, of thin stratified regions [21]. The TFT returns an approximated solution to Maxwell's equations, similarly to Fresnel's equations, for two semi-infinite media enclosing a region of the space, hereby identified as a thin film, with a previously unspecified composition [9, 21, 37]. The optical properties of the thin film, under the excess field formalism, are determined by introducing a polarization and magnetization fields on the film through constitutive equations [21]. For example, the thin film can model an ensemble of nanoparticles given their material and density within the array [38, 39]. The excess field approach modifies Maxwell's equations at the interfaces of the system, thus determining explicitly the discontinuities of the electromagnetic field at the boundaries of the thin film.

To study the reflection and transmission of an electromagnetic plane wave illuminating an arbitrary thin film, of thickness d, centered at $z = 0$ in between two media characterized by their bulk dielectric functions or refractive indices $\varepsilon^{\gtrless} = (n^{\gtrless})^2$, it is proposed that the total electric field $\mathbf{E}(\mathbf{r})$ is decomposed as follows [21, 39]

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}^{>}(\mathbf{r})\Theta(z) + \mathbf{E}^{<}(\mathbf{r})\Theta(-z) + \mathbf{E}^{\text{excess}}(\mathbf{r}), \tag{1.15}$$

where the symbols $>$ and $<$ denote the semiinfinite media above and below the thin film, respectively. In Eq. (1.15) the time harmonic dependency in ω is implicitly included and $\Theta(z)$ is the Heaviside step function; the electric fields $\mathbf{E}^{\gtrless}$ are considered to be radiation fields [21]. The term $\mathbf{E}^{\text{excess}}(\mathbf{r})$ is the excess electric field and its contribution is non-zero in the neighborhood of the thin film since $\mathbf{E}(\mathbf{r}) \to \mathbf{E}^{\gtrless}(\mathbf{r})$ when $z \to \pm\infty$. Thus, the whole effect of the thin film in the electromagnetic properties of the system can be described by considering the total excess electric field in the film $\mathbf{E}^{\text{film}}(\mathbf{r}, z = 0)$ in the limit when $d \to 0$, thus, yielding the expression [21, 39]

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}^{>}(\mathbf{r})\Theta(z) + \mathbf{E}^{<}(\mathbf{r})\Theta(-z) + \mathbf{E}^{\text{film}}(\mathbf{r}_{\parallel})\delta(z), \tag{1.16}$$

with $\delta(z)$ the Dirac delta function and

$$\mathbf{E}^{\text{film}}(\mathbf{r}_{\parallel}) = \int_{-\infty}^{\infty} \mathbf{E}^{\text{excess}}(\mathbf{r})\,\mathrm{d}z = \left(\int_{-\infty}^{\infty} \mathbf{E}^{\text{excess}}(\mathbf{r}) \exp(ik_z z)\,\mathrm{d}z \right)\bigg|_{k_z=0}, \tag{1.17}$$

where $\mathbf{r}_{\parallel}$ is a vector evaluated at the plane $z = 0$. In Eq. (1.17) it is explicitly shown that $\mathbf{E}^{\text{film}}(\mathbf{r}_{\parallel})$ equals the Fourier transform of the excess electric field in the z variable evaluated at $k_z = 0$. The same decomposition employed in Eqs. (1.15)–(1.17) for $\mathbf{E}(\mathbf{r})$ applies to the magnetic field $\mathbf{B}(\mathbf{r})$, as well as any other scalar or vector field relevant to the system [39], for example, the electric displacement field $\mathbf{D}(\mathbf{r})$, the $\mathbf{H}(\mathbf{r})$ field, and the external charge and external current densities, $\rho_{\text{ext}}(\mathbf{r})$ and $\mathbf{J}_{\text{ext}}(\mathbf{r})$. By substituting such definitions of the fields into Maxwell's equations, the following expressions for the excess quantities are obtained [21]

$$\nabla \cdot \mathbf{D}^{\text{excess}}(\mathbf{r}) + \left[D_{\perp}^{>}(\mathbf{r}_{\parallel}) - D_{\perp}^{<}(\mathbf{r}_{\parallel}) \right] \delta(z) = \rho_{\text{ext}}^{\text{excess}}(\mathbf{r}), \tag{1.18a}$$

$$\nabla \cdot \mathbf{B}^{\text{excess}}(\mathbf{r}) + \left[B_{\perp}^{>}(\mathbf{r}_{\parallel}) - B_{\perp}^{<}(\mathbf{r}_{\parallel}) \right] \delta(z) = 0, \tag{1.18b}$$

$$\nabla \times \mathbf{E}^{\text{excess}}(\mathbf{r}) + \hat{\mathbf{e}}_{\perp} \times \left[\mathbf{E}_{\parallel}^{>}(\mathbf{r}_{\parallel}) - \mathbf{E}_{\parallel}^{<}(\mathbf{r}_{\parallel}) \right] \delta(z) = +i\omega \mathbf{B}^{\text{excess}}(\mathbf{r}), \tag{1.18c}$$

$$\nabla \times \mathbf{H}^{\text{excess}}(\mathbf{r}) + \hat{\mathbf{e}}_{\perp} \times \left[\mathbf{H}_{\parallel}^{>}(\mathbf{r}_{\parallel}) - \mathbf{H}_{\parallel}^{<}(\mathbf{r}_{\parallel}) \right] \delta(z) = \mathbf{J}_{\text{ext}}^{\text{excess}}(\mathbf{r}) - i\omega \mathbf{D}^{\text{excess}}(\mathbf{r}), \tag{1.18d}$$

where it was considered that the fields in the semi-infinite media are solution to Maxwell's equations and that the step and delta functions are related as $\mathrm{d}\Theta(\pm z)/\mathrm{d}z = \pm\delta(z)$. The subscript $\parallel$ corresponds to the parallel components of the fields relative to the thin film, $\perp$ to their normal components to the film; $\hat{\mathbf{e}}_{\perp}$ is the unitary vector in this direction. The terms in between brackets in Eqs. (1.18) are the boundary conditions of the electromagnetic fields evaluated at the thin film and they are modified from the known case of a sharp interface —which yield Fresnel's equations— by the excess quantities [21].

The boundary conditions of the electromagnetic fields in the TFT formalism can be determined by calculating the Fourier transform of Eqs. (1.18) in the z variable and then considering the particular case of $k_z = 0$, so that all excess quantities are interchanged for their total contribution according to Eq. (1.17). Additionally, total excess polarization $\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel})$ and magnetization $\mathbf{M}^{\text{film}}(\mathbf{r}_{\parallel})$ are defined in terms of the total excess electromagnetic fields, thus the boundary conditions in the absence of external sources —$\mathbf{J}_{\text{ext}}(\mathbf{r}) = \mathbf{0}$ and $\rho_{\text{ext}}(\mathbf{r}) = 0$— can be written as [21, 39]

$$\left[D_\perp^>(\mathbf{r}_\parallel) - D_\perp^<(\mathbf{r}_\parallel)\right] = -\nabla_\parallel \cdot \mathbf{P}_\parallel^{\text{film}}(\mathbf{r}_\parallel), \tag{1.19a}$$

$$\left[B_\perp^>(\mathbf{r}_\parallel) - B_\perp^<(\mathbf{r}_\parallel)\right] = -\nabla_\parallel \cdot \mathbf{M}_\parallel^{\text{film}}(\mathbf{r}_\parallel), \tag{1.19b}$$

$$\left[\mathbf{E}_\parallel^>(\mathbf{r}_\parallel) - \mathbf{E}_\parallel^<(\mathbf{r}_\parallel)\right] = -i\omega\hat{\mathbf{e}}_\perp \times \mathbf{M}_\parallel^{\text{film}}(\mathbf{r}_\parallel) - \nabla_\parallel P_\perp^{\text{film}}(\mathbf{r}_\parallel), \tag{1.19c}$$

$$\left[\mathbf{H}_\parallel^>(\mathbf{r}_\parallel) - \mathbf{H}_\parallel^<(\mathbf{r}_\parallel)\right] = +i\omega\hat{\mathbf{e}}_\perp \times \mathbf{P}_\parallel^{\text{film}}(\mathbf{r}_\parallel) - \nabla_\parallel M_\perp^{\text{film}}(\mathbf{r}_\parallel), \tag{1.19d}$$

where $\nabla_\parallel$ and $\nabla_\parallel\cdot$ are the bidimensional gradient and divergence operators on the thin film, respectively, and [21]

$$\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel) = \mathbf{D}_\parallel^{\text{film}}(\mathbf{r}_\parallel) - \varepsilon_0 E_\perp^{\text{film}}(\mathbf{r}_\parallel)\hat{\mathbf{e}}_\perp, \tag{1.20}$$

$$\mathbf{M}^{\text{film}}(\mathbf{r}_\parallel) = \frac{1}{\mu_0}\mathbf{B}_\parallel^{\text{film}}(\mathbf{r}_\parallel) - H_\perp^{\text{film}}(\mathbf{r}_\parallel)\hat{\mathbf{e}}_\perp, \tag{1.21}$$

with ε_0 (μ_0) the electric permittivity (magnetic permeability) of the vacuum. It is noteworthy that the definition of $\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel)$ in Eq. (1.20) is not just a direct identification of the procedure from Eqs. (1.18) to Eqs. (1.19) but it is based on the non-continuity of $\mathbf{D}_\parallel(\mathbf{r})$ and $E_\perp(\mathbf{r})$ across boundaries, which give rise to an excess polarization on the film unlike $D_\perp(\mathbf{r})$ and $\mathbf{E}_\parallel(\mathbf{r})$, whose continuity do not have such an effect on the system [21, 39]. An analogous argument is performed on the definition of the magnetization field in Eq. (1.21).

The boundary conditions for the electromagnetic fields, given in Eq. (1.19), reduce to Fresnel's equations when the film represents a planar sharp interface between two linear homogeneous media by setting $\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel) = \mathbf{M}^{\text{film}}(\mathbf{r}_\parallel) = \mathbf{0}$ [21]. To introduce a thin film with a different, but known, structure, the total excess polarization and magnetization [Eqs. (1.20) and (1.21)] can be described by constitutive equations as an expansion to different orders of the electromagnetic fields [21]. The particular case where the thin film corresponds to a collection of identical plasmonic scatterers, with a size where no magnetic response is expected, the general constitutive relations to Eqs. (1.20) and (1.21) can be expressed to a linear order of any component of total excess electric field or the total excess displacement field [38, 39]. Since $\mathbf{E}^{\text{film}}(\mathbf{r}_\parallel)$ and $\mathbf{D}^{\text{film}}(\mathbf{r}_\parallel)$ are linearly dependent, the total excess polarization and magnetization can be expressed as [21, 39]

$$\mathbf{P}^{\text{film}}(\mathbf{r}_\parallel) = \gamma\,\langle\mathbf{E}_\parallel^{\text{film}}(\mathbf{r}_\parallel)\rangle + \beta\,\langle D_\perp^{\text{film}}(\mathbf{r}_\parallel)\rangle\hat{\mathbf{e}}_\perp \qquad \text{and} \qquad \mathbf{M}^{\text{film}}(\mathbf{r}_\parallel) = \mathbf{0} \tag{1.22}$$

where γ and β are known as the film susceptibilities, which encodes the physical properties of the ensemble of scatterers, and where the operator $\langle\cdot\rangle$ corresponds to the average on the film given as $\langle a^{\text{film}}(\mathbf{r}_\parallel)\rangle = (1/2)[a^>(\mathbf{r}_\parallel) + a^<(\mathbf{r}_\parallel)]$, for either a scalar or vector quantity $a^{\gtrless}(\mathbf{r})$ [21, 39].

The TFT employs the excess field formalism to determine the optical response of a metasurface embedded in a matrix and supported on a substrate, as shown in Fig. 1.2, both non-absorbing nor magnetic ($\mu^{\gtrless} = \mu_0$), by calculating amplitude reflection r_{TFT} and transmission t_{TFT} coefficients from Eqs. (1.19) and (1.22), considering that the thin film consists in a disordered ensemble of identical plasmonic spherical particles. For a configuration of internal illumination, the electric field below the thin film corresponds to the superposition of an incident and a reflected electromagnetic plane waves $\mathbf{E}^{\text{inc}}(\mathbf{r})$ and $\mathbf{E}_{\text{r}}(\mathbf{r})$ traveling in the $\hat{\mathbf{k}}^{\text{inc}}$ and $\hat{\mathbf{k}}^{\text{r}}$ directions,

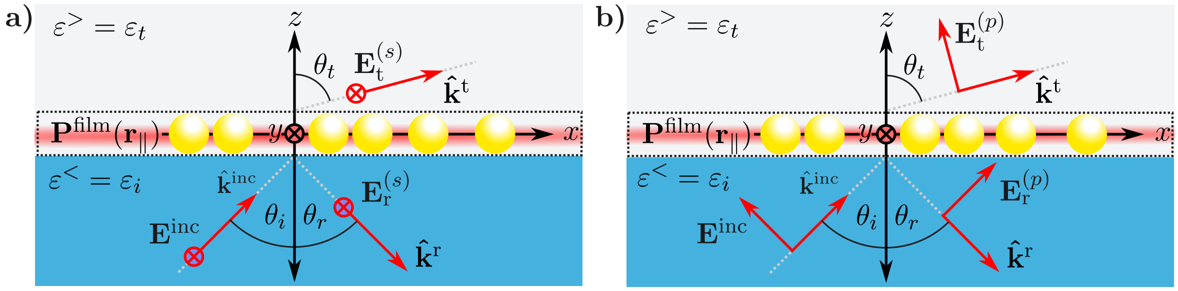

Fig. 1.2: Diagram of the reflection of transmission of **a)** an s polarized and **b)** a p polarized electromagnetic plane wave $\mathbf{E}^{\mathrm{inc}}$ traveling in the $\hat{\mathbf{k}}^{\mathrm{inc}}$ direction at an angle of incidence θ_i that illuminates a collection of identical spherical particles from a medium characterized by the dielectric function $\varepsilon^{<} = \varepsilon_i$ (blue region) into another medium characterized by $\varepsilon^{>} = \varepsilon_t$ (grey region). The thin film (dashed line) models the electromagnetic response of the ensemble of particles embedded in the transmission medium through the total excess polarization $\mathbf{P}^{\mathrm{film}}(\mathbf{r}_{\parallel})$ (represented by the red blurred area) at $z = 0$. The reflected $\mathbf{E}_{\mathrm{r}}^{(j)}$ and transmitted electric field $\mathbf{E}_{\mathrm{t}}^{(j)}$ for a j polarization state travel at the $\hat{\mathbf{k}}_{\mathrm{r}}$ and $\hat{\mathbf{k}}_{\mathrm{t}}$ direction, respectively, that is, at the reflection θ_r and transmission angle θ_t.

respectively, within the substrate characterized by the dielectric function $\varepsilon^{<} = \varepsilon_i = n_i^2$, while above the electric field is given by the plane wave $\mathbf{E}^{\mathrm{t}}(\mathbf{r})$ traveling in the $\hat{\mathbf{k}}^{\mathrm{t}}$ direction in the matrix with $\varepsilon^{>} = \varepsilon_t = n_t^2$. To determine the amplitude coefficients under the TFT the cases of an s and a p polarized incident electromagnetic wave are treated separately.

In general, when an electromagnetic plane wave with an angular frequency ω travels from a substrate to a thin film in the direction $\hat{\mathbf{k}}^{\mathrm{inc}}$ at an angle θ_i, relative to the normal direction to the thin film, the reflected (transmitted) plane wave does it in the direction of $\hat{\mathbf{k}}^{\mathrm{r}}$ ($\hat{\mathbf{k}}^{\mathrm{t}}$) at the angle θ_r (θ_t). Thus, the wave vectors of the electromagnetic plane waves are

$$\mathbf{k}^{\mathrm{inc}} = n_i \frac{\omega}{c}(\sin\theta_i\,\hat{\mathbf{e}}_x + \cos\theta_i\,\hat{\mathbf{e}}_\perp), \tag{1.23a}$$

$$\mathbf{k}^{\mathrm{r}} = n_i \frac{\omega}{c}(\sin\theta_r\,\hat{\mathbf{e}}_x - \cos\theta_r\,\hat{\mathbf{e}}_\perp), \tag{1.23b}$$

$$\mathbf{k}^{\mathrm{t}} = n_t \frac{\omega}{c}(\sin\theta_t\,\hat{\mathbf{e}}_x + \cos\theta_t\,\hat{\mathbf{e}}_\perp), \tag{1.23c}$$

where $\hat{\mathbf{e}}_\perp = \hat{\mathbf{e}}_z$ and c is the speed of light. For an s polarized electromagnetic plane wave [see Fig. 1.2a)], the electric field above and below the thin film is

$$\mathbf{E}^{<}(\mathbf{r}) = \left[E^{\mathrm{inc}} \exp\left(i\mathbf{k}^{\mathrm{inc}}\cdot\mathbf{r}\right) + E_{\mathrm{r}}^{(s)} \exp(i\mathbf{k}^{\mathrm{r}}\cdot\mathbf{r})\right]\hat{\mathbf{e}}_y, \tag{1.24a}$$

$$\mathbf{E}^{>}(\mathbf{r}) = E_{\mathrm{t}}^{(s)} \exp\left(i\mathbf{k}^{\mathrm{t}}\cdot\mathbf{r}\right)\hat{\mathbf{e}}_y, \tag{1.24b}$$

while for a p polarized plane wave, the electric field is given by

$$\begin{aligned}\mathbf{E}^{<}(\mathbf{r}) =& \left[-E^{\mathrm{inc}}\cos\theta_i \exp\left(i\mathbf{k}^{\mathrm{inc}}\cdot\mathbf{r}\right) + E_{\mathrm{r}}^{(p)}\cos\theta_r \exp(i\mathbf{k}^{\mathrm{r}}\cdot\mathbf{r})\right]\hat{\mathbf{e}}_x + \\ &+ \left[E^{\mathrm{inc}}\sin\theta_i \exp\left(i\mathbf{k}^{\mathrm{inc}}\cdot\mathbf{r}\right) + E_{\mathrm{r}}^{(p)}\sin\theta_r \exp(i\mathbf{k}^{\mathrm{r}}\cdot\mathbf{r})\right]\hat{\mathbf{e}}_\perp,\end{aligned} \tag{1.25a}$$

$$\mathbf{E}^{>}(\mathbf{r}) = \left[-E_{\mathrm{t}}^{(p)}\cos\theta_t \exp\left(i\mathbf{k}^{\mathrm{t}}\cdot\mathbf{r}\right)\right]\hat{\mathbf{e}}_x + \left[-E_{\mathrm{t}}^{(p)}\sin\theta_t \exp\left(i\mathbf{k}^{\mathrm{t}}\cdot\mathbf{r}\right)\right]\hat{\mathbf{e}}_\perp, \tag{1.25b}$$

as it can be seen in Fig. 1.2b). In Eqs. (1.24) and (1.25), the term $E_{\xi}^{(j)}$ is the magnitude of either the reflected ($\xi = \mathrm{t}$) or transmitted ($\xi = \mathrm{t}$) electric field considering the j polarization state; for either case E^{inc} is the magnitude of the incident electric field. Additionally, the incident $\mathbf{B}^{\mathrm{inc}}(\mathbf{r})$, and the reflected and transmitted magnetic fields $\mathbf{B}_{\xi}^{(j)}(\mathbf{r})$ for both polarizations can be determined by substituting Eqs. (1.24) and (1.25) into Faraday-Lenz's Law, while the displacement field and the H field are calculated via the bulk contitutive relations $\mathbf{D} = \varepsilon\mathbf{E}(\mathbf{r})$ and $\mathbf{H}(\mathbf{r}) = \mathbf{B}(\mathbf{r})/\mu_0$ considering each medium linear, homogeneous and isotropic. Prior calculation of the total polarization in the film $\mathbf{P}^{\mathrm{film}}(\mathbf{r}_{\parallel})$ it is noteworthy to show that forcing the proposed electric fields in Eqs. (1.24) and (1.25) to follow the boundary conditions shown in Eqs. (1.19) yields the continuity of the parallel component of the wave vector $\mathbf{k}_{\parallel} = (\omega/c)n_i \sin\theta_i \hat{\mathbf{e}}_x = (\omega/c)n_t \sin\theta_t\, \hat{\mathbf{e}}_x$, therefore stating that even with the presence of the thin film, the reflection and Snell's law are valid [21, 39]. Under this consideration, $\mathbf{P}^{\mathrm{film}}(\mathbf{r}_{\parallel})$ is obtained by introducing Eqs. (1.24) and (1.25) into Eq. (1.22) yielding

$$\mathbf{P}_{\parallel}^{(s),\mathrm{film}}(\mathbf{r}_{\parallel}) = \frac{\gamma}{2}\Big(E^{\mathrm{inc}} + E_{\mathrm{r}}^{(s)} + E_{\mathrm{t}}^{(s)}\Big)\exp\Big(\mathbf{k}_{\parallel}\cdot\mathbf{r}\Big)\hat{\mathbf{e}}_x, \tag{1.26a}$$

$$P_{\perp}^{(s),\mathrm{film}}(\mathbf{r}_{\parallel}) = 0, \tag{1.26b}$$

for s polarization and

$$\mathbf{P}_{\parallel}^{(p),\mathrm{film}}(\mathbf{r}_{\parallel}) = \frac{\gamma}{2}\Big[\Big(E_{\mathrm{r}}^{(p)} - E^{\mathrm{inc}}\Big)\cos\theta_i - E_{\mathrm{t}}^{(p)}\cos\theta_t\Big]\exp\Big(\mathbf{k}_{\parallel}\cdot\mathbf{r}\Big)\hat{\mathbf{e}}_x, \tag{1.27a}$$

$$P_{\perp}^{(p),\mathrm{film}}(\mathbf{r}_{\parallel}) = \frac{\beta}{2}\Big[\varepsilon_i\Big(E^{\mathrm{inc}} + E_{\mathrm{r}}^{(p)}\Big)\sin\theta_i + \varepsilon_t E_{\mathrm{t}}^{(p)}\sin\theta_t\Big]\exp\Big(\mathbf{k}_{\parallel}\cdot\mathbf{r}\Big), \tag{1.27b}$$

for p polarization.

As stated above, the TFT returns expressions for reflection and transmission amplitude coefficients. These coefficients are obtained for an s polarized incident plane wave by solving the system of equations obtained by substituting Eqs. (1.23), (1.24) and (1.26) into the boundary conditions in Eq. (1.19), yielding

$$E^{\mathrm{inc}} + E_{\mathrm{r}}^{(s)} = E_{\mathrm{t}}^{(s)}, \tag{1.28a}$$

$$\Big(n_i\cos\theta_i + \frac{i}{2}\frac{\omega}{c}\gamma\Big)E^{\mathrm{inc}} - \Big(n_i\cos\theta_i - \frac{i}{2}\frac{\omega}{c}\gamma\Big)E_{\mathrm{r}}^{(s)} = \Big(n_t\cos\theta_t - \frac{i}{2}\frac{\omega}{c}\gamma\Big)E_{\mathrm{t}}^{(s)}. \tag{1.28b}$$

Analogously, the system of equations for a p polarized incident plane wave obtained from Eqs. (1.19), (1.23) (1.25) and (1.27) is

$$\Big(\cos\theta_i + ik_{\parallel}\frac{\beta}{2}\varepsilon_i\sin\theta_i\Big)E^{\mathrm{inc}} - \Big(\cos\theta_i - ik_{\parallel}\frac{\beta}{2}\varepsilon_i\sin\theta_i\Big)E_{\mathrm{r}}^{(p)} = \Big(\cos\theta_t - ik_{\parallel}\frac{\beta}{2}\varepsilon_t\sin\theta_t\Big)E_{\mathrm{t}}^{(p)}, \tag{1.29a}$$

$$\Big(n_i + ik_{\parallel}\frac{\gamma}{2}\cos\theta_i\Big)E^{\mathrm{inc}} + \Big(n_i - ik_{\parallel}\frac{\gamma}{2}\cos\theta_i\Big)E_{\mathrm{r}}^{(p)} = \Big(n_t - ik_{\parallel}\frac{\gamma}{2}\cos\theta_t\Big)E_{\mathrm{t}}^{(p)}, \tag{1.29b}$$

with $k_{\parallel} = (\omega/c)n_i\sin\theta_i = (\omega/c)n_t\sin\theta_t$. Thus, the reflection and transmission amplitude coefficients given by the solution to Eqs. (1.28) for $\xi_{\mathrm{TFT}}^{(s)} = E_{\xi}^{(s)}/E^{\mathrm{inc}}$ are [9, 21, 39]

$$r_{\mathrm{TFT}}^{(s)}(\theta_i) = \frac{n_i\cos\theta_i - n_t\cos\theta_t + i(\omega/c)\gamma}{n_i\cos\theta_i + n_t\cos\theta_t - i(\omega/c)\gamma}, \tag{1.30a}$$

$$t^{(s)}_{\mathrm{TFT}}(\theta_i) = \frac{2n_i\cos\theta_i}{n_i\cos\theta_i + n_t\cos\theta_t - i(\omega/c)\gamma}, \tag{1.30b}$$

while for $\xi^{(p)}_{\mathrm{TFT}} = E^{(p)}_{\xi}/E^{\mathrm{inc}}$, obtained by solving Eqs. (1.29), the amplitude coefficients are [9, 21, 39]

$$r^{(p)}_{\mathrm{TFT}}(\theta_i) = \frac{\kappa_-(\omega) - i(\omega/c)\gamma\cos\theta_i\cos\theta_t + i(\omega/c)n_i n_t\varepsilon_i\beta\sin^2\theta_i}{\kappa_+(\omega) - i(\omega/c)\gamma\cos\theta_i\cos\theta_t - i(\omega/c)n_i n_t\varepsilon_i\beta\sin^2\theta_i}, \tag{1.31a}$$

$$t^{(p)}_{\mathrm{TFT}}(\theta_i) = \frac{n_i\cos\theta_i\Big(1-(\omega/2c)^2\varepsilon_i\gamma\beta\sin^2\theta_i\Big)}{\kappa_+(\omega) - i(\omega/c)\gamma\cos\theta_i\cos\theta_t - i(\omega/c)n_i n_t\varepsilon_i\beta\sin^2\theta_i}, \tag{1.31b}$$

$$\kappa_\pm(\omega) = \Big(n_t\cos\theta_i \pm n_i\cos\theta_t\Big)\left[1-\left(\frac{\omega}{2c}\right)^2\varepsilon_i\gamma\beta\sin^2\theta_i\right]. \tag{1.31c}$$

To completely describe the optical properties of a metasurface of identical plasmonic scatterers supported by a substrate with Eqs. (1.30) and (1.31), the film susceptibilities γ and β defined in Eq. (1.22) must be specified. When considering spherical scatterers of radius a embedded in the transmission medium and supported on a planar sharp interface with the incidence medium, the electromagnetic response of the ensemble can be modeled by electric point dipoles and their images if the condition $n_t a\omega/c \ll 1$ is fulfilled [39]. Under these considerations in addition to setting the thickness of the thin film modeling the ensemble of particles as $2a$, the film susceptibilities are given by [21, 38, 39]

$$\gamma = \frac{2}{3}\frac{\Theta}{V_{\mathrm{sp}}}\left(\frac{\alpha_{\mathrm{sp}}}{1+A\alpha_{\mathrm{sp}}}\right) \qquad \text{and} \qquad \beta = \frac{2}{3}\frac{\Theta}{\varepsilon_i^2 V_{\mathrm{sp}}}\left(\frac{\alpha_{\mathrm{sp}}}{1+2A\alpha_{\mathrm{sp}}}\right) \tag{1.32}$$

where Θ is the surface coverage of the particles ensemble on the substrate and where α_{sp} and A are the polarizability of the spherical particles and the image strength, respectively, defined as

$$\alpha_{\mathrm{sp}} = 3\varepsilon_t V_{\mathrm{sp}}\left(\frac{\varepsilon_{\mathrm{sp}}-\varepsilon_t}{\varepsilon_{\mathrm{sp}}+2\varepsilon_t}\right) \qquad \text{and} \qquad A = \frac{\varepsilon_i - \varepsilon_t}{\varepsilon_i+\varepsilon_t}, \tag{1.33}$$

with $\varepsilon_{\mathrm{sp}}$ the dielectric function of the spheres and V_{sp} their volume. It is noteworthy that the expressions of r_{TFT} and t_{TFT} shown in the Eqs. (1.30) and (1.31) describe the optical properties of the ensemble when the particles are embedded in the transmission medium. By interchanging $\varepsilon_t \leftrightarrow \varepsilon_i$ and $n_t \leftrightarrow n_i$ in Eqs. (1.30)–(1.33), the amplitude coefficients describe the scenario when the particles are embedded in the medium of incidence [21].

1.3 Experimental Setup

In order to characterize and analyze the optical response of a metasurface of spherical gold nanoparticles (AuNPs) with a disordered spatial distribution supported on a glass substrate, the extinction and reflectivity spectra of the metasurface are measured. In Fig. 1.3 the setup used for the extinction measurements is shown, while in Fig. 1.4 it is presented the setup for the reflectivity measurements.

The extinction measurements, as schematized in Fig. 1.3a), are obtained by illuminating the metasurface with a white light source (LS) at normal incidence at an external configuration, that is, from air into the substrate. The light is collimated with a lens (L) before the AuNPs metasurface and then the transmitted light is collected with a glass fiber and sent to an spectrometer (S). An image of the experimental setup is presented in Fig. 1.3b) where each component of the setup is signalized. To perform the extinction measurements, it was employed a halogen lamp (Mikropack HL-2000) as the white light source and a StellarNet(R) BLUE-Wave Spectrometer with spectral range between 350 nm and 1,150 nm.

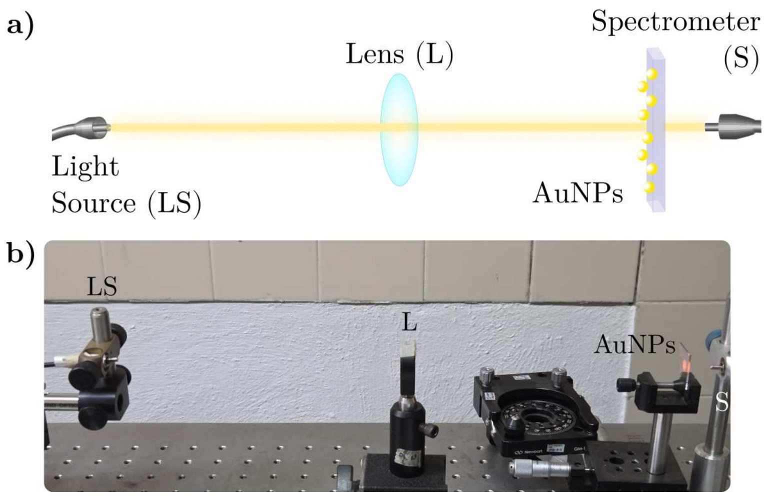

Fig. 1.3: **a)** Schematic representation and **b)** photograph of the experimental setup to perform extinction spectra measurements. The setup consists in a white light source (LS), whose light beam is collimated by a lens (L) before illuminating a metasurface of AuNPs at normal incidence. The transmitted light through the metasurface is collected into a spectrometer (S).

The reflectivity spectra is measured in an internal illumination configuration with a white light beam from a Xenon lamp (ThorLabs SLS201L) as a light source (LS). In Fig. 1.4a) it is shown the schematic representation of the experimental setup presented in Fig. 1.4b), where it can be seen that the light beam passes through a lens (L) and a linear polarizer (LP) before striking into a planar face of a right-angle prism (P) supporting the metasurface. The reflected beam from the metasurface is then collected by a second lens (L) and directed into a spectrometer (S): Ocean Optics HR4000 Spectrometer. To perform the reflectivity measurements, the prism (BK7 glass) and the substrate of the metasurface (Corning glass) were optically coupled by index-matching their materials with a microscope immersion oil with a refractive index of $n = 1.518$.

A scheme of the optically coupled prism and the metasurface is shown in Fig. 1.5a) and an overlap of the scheme and the actual system in 1.5b). The red arrows correspond to the optical path of the light beam through the prism into the AuNPs, the gray line to the immersion oil index-matching the metasurface and the prism, and the dashed lines represent the normal direction to the planar faces of the prism. The light beam travels into the prism at an angle θ_g, experimentally set on a goniometer holding the prism as shown in Fig. 1.5b), which determines the angle of incidence θ_i at which the AuNPs are illuminated. The relation between θ_g and θ_i is

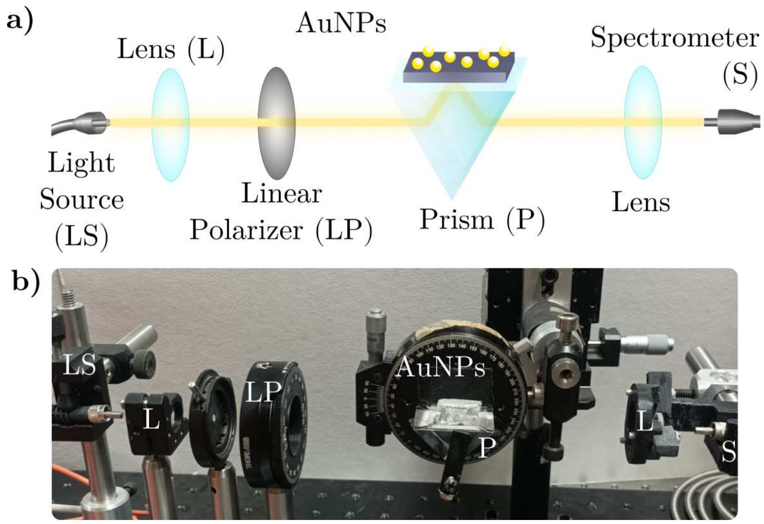

Fig. 1.4: a) Schematic representation and **b)** photograph of the experimental setup to perform reflectivity spectra measurements. The setup consists in a white light source (LS), whose light beam passes through a lens (L) and a linear polarizer (LP) before striking a glass prism (P). The metasurface of AuNPs is coupled with an immersion oil so the AuNPs are illumited from inside the prism at the desired angle of incidence. The reflected light is collected with a lens (L) and directed into a spectrometer (S).

given by

$$\theta_g(\theta_i) = \sin^{-1}\left[\frac{n_{\mathrm{Prism}}}{n_{\mathrm{Air}}}\sin\left(\theta_i - \frac{\pi}{4}\right)\right],$$

where n_{Air} is the refractive index of the air (typically set equal to the unity) and n_{Prism} that of the prism (typically set equal to 1.5). Due to the index matching, the refraction between the prism and the matasurfaces's substrate is neglected.

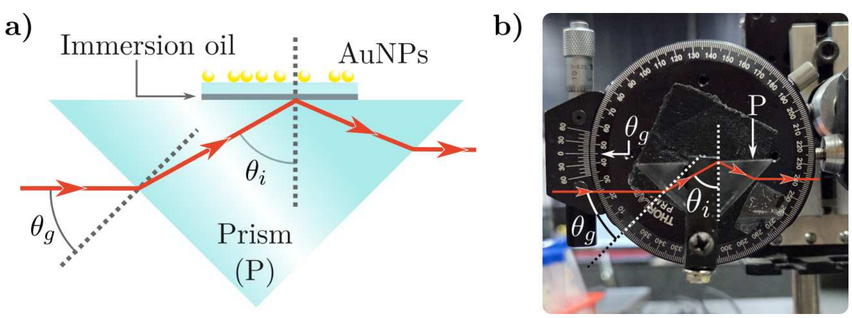

Fig. 1.5: a) Schematic representation of the right-angle prism (P) mounted in the reflectivity measurement setup described in Fig. 1.4 and **b)** a photograph of the prism attached to a goniometer used to control the angle of incidence θ_i of a light beam on the metasurface of AuNPs (not shown in the photograph). A light beam (red line) entering the prism at an angle θ_g, measured with the goniometer scale, is refracted inside the prism according to Snell's law. This refraction determines θ_i at the interface between the prism and the metasurface's substrate, which are optically coupled with an immersion oil layer (gray solid line). The dashed gray lines correspond to the normal direction to any of the planar faces of the prism.

In addition to the reflectivity measurements, ellipsometric technics can be achived with the setup in Fig. 1.4 after the introduction of a single optical element in the optical path of the incident light. In particular, the phase change of the reflected light by a AuNPs metasurface can be determine with the spectra of the ellipsometric parameters ψ and $\Delta\phi$ defined as

$$r^{(p)}/r^{(s)} = \tan\psi \exp(i\Delta\phi), \tag{1.34}$$

where $r^{(j)}$ is the reflection amplitude coefficient for the j polarization state. The $\Delta\phi$ parameter corresponds to the phase difference between the reflected p and s polarized components of light. The experimental determination of $\Delta\phi$ can be performed with the experimental setup shown in Fig. 1.4 by placing a birefringent crystal between the first linear polarizer and the prism supporting the metasurface with the crystal's optical axis oriented at 45° with respect to polarizer's axis as described in Ref. [9].

Chapter 2

Progress in Non-destructive Characterization

In this Chapter the advances on the identification of substrate effects in the optical properties of metasurfaces are presented. These results consist in the comparison between measurements performed by collaborators at the Instituto Nacional de Astrofísica, Óptica y Electrónica (INAOE) of extinction and reflectivity spectra —with the setups presented in Section 1.3— of metasurfaces with theoretical and numerical results. The analyzed metasurfaces were fabricated at INAOE by dewetting gold thin films yielding disordered ensembles of spherical gold nanoparticles (AuNP) supported on a Corning glass substrate. In Section 2.1 the experimental extinction and reflectivity spectra of two metasurfaces were contrasted with theoretical results obtained with the Coherent Scattering Model (CSM) alongside Finite Element Method (FEM) simulations performed in COMSOL Multiphysics$^{\text{TM}}$ ver. 6.2 in order to quantify independently the effect of the multiple scattering, the particle-substrate interaction and a possible partial embedding of the AuNPs into the substrate. To further characterize the partial embedding in metasurfaces, in Section 2.2 there are presented the first performed measurements for two sets of metasurfaces expected to have different incrustation degree into the substrate. While analyzes of the samples with Scanning Electron Microscopy (SEM) or Atomic Force Microscopy (AFM) images are needed to deepen into this topic, first optical characterizations —which I conducted during a short fellowship at INAOE— aimed to validate the partial embedding are presented in Section 2.1.

2.1 Substrate Effects on a Plasmonic Metasurface

The metasurfaces analyzed in this Section were synthesized at INAOE by collaborators by thermal dewetting from gold films with initial thicknesses $d = (3.0 \pm 0.2)$ nm and $d = (5.0 \pm 0.2)$ nm which were held inside a Caisa DTT434 oven at 550°C for 3 hours. SEM images of each metasurface are shown in the background of Figs. 2.1a) and 2.1b) for the $d = 3$ nm and $d = 5$ nm samples, respectively. The orange curves in the foreground correspond to the histograms of the AuNPs classified by their equivalent radius, while the red and green curves correspond to the Gaussian-fitted distributions of the histograms, from which the mean radius μ and the standard deviation σ of the size distribution of each sample were determined. The sample with initial thickness of $d = 3$ nm, shown in Fig. 2.1a), is characterized with parameters $\mu = (5.78 \pm 0.06)$ nm, $\sigma = (1.89 \pm 0.06)$ nm, and surface coverage $\Theta = 0.20 \pm 0.01$. In contrast, the sample with initial thickness of $d = 5$ nm, shown in Fig. 2.1b), is characterized with $\mu = (7.30 \pm 0.05)$ nm,

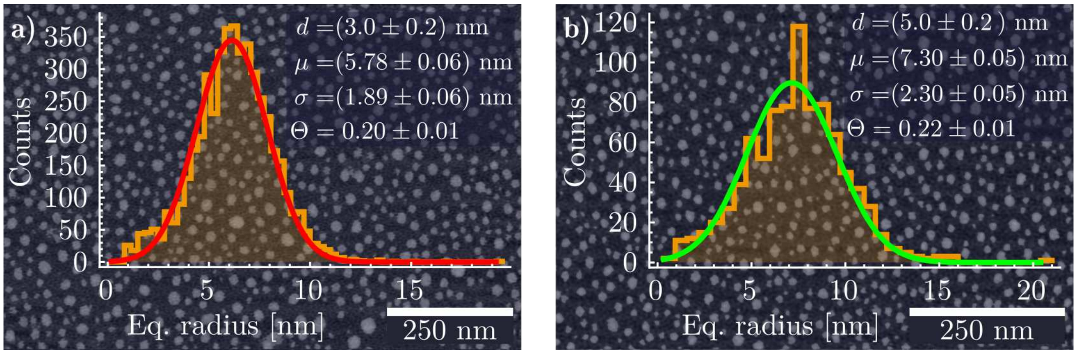

Fig. 2.1: SEM image and histogram of the equivalent radius of the AuNPs in a metasurface fabricated from a gold film with an initial thickness **a)** $d = (3.0 \pm 0.2)$ nm and **b)** $d = (5.0 \pm 0.02)$ nm, both showing a normal distribution of the equivalent radius of the AuNPs with a mean μ, standard deviation σ, and surface coverage Θ.

$\sigma = (2.30 \pm 0.05)$ nm, and $\Theta = 0.22 \ \pm \ 0.01$.

In Fig. 2.2a) the experimental extinction efficiencies (scale on the left axis for orange lines) at normal incidence, as a function of the incident wavelength for the metasurface characterized by $\mu = (5.78 \pm 0.06)$ nm in the top panel and by $\mu = (7.30 \pm 0.05)$ nm in the bottom panel. These measurements are contrasted with the theoretical extinction efficiencies (scale on the right axis) of an isolated AuNP, calculated through standard Mie Theory, with radii $a = 5.78$ nm (red lines) and $a = 7.30$ nm (green lines). The calculations consider a matrix of either air (long dashed lines) with $n_{\mathrm{m}} = 1$ or glass (short dashed lines) with $n_{\mathrm{m}} = 1.5$. The refractive index used for the AuNP $n_{\mathrm{NP}}(\omega)$ was obtained from the experimental data reported by Johnson and Christy [40], modified with a size correction [41] to the Drude contribution of the complex-valued dielectric function $\varepsilon_{\mathrm{NP}}(\omega) = n^2_{\mathrm{NP}}(\omega)$.

From the experimental extinction efficiencies, a resonance is observed at a wavelength of approximately 542 nm (546 nm) for the metasurfaces characterized by a nominal mean radius of $\mu = 5.78$ nm ($\mu = 7.30$ nm), shown in the upper (lower) panel. The theoretical extinction efficiencies for isolated AuNPs are found to be maximized at around 510 nm when $n_{\mathrm{m}} = 1$ and at approximately 533 nm when $n_{\mathrm{m}} = 1.5$, for both chosen values of the nanoparticle radius a. These wavelengths correspond to the excitation of the electric dipolar Localized Surface Plasmon Resonance (LSPR). When the extinction spectra of the metasurfaces in Fig. 2.2a) are compared to those of the corresponding isolated AuNPs, a redshift in the resonance wavelength of the metasurfaces is observed. For instance, if the AuNPs in the metasurfaces were treated as independent scatterers, embedded in air and perfectly supported on glass, a redshift of approximately 36 nm would be expected. A similar analysis can be performed considering the AuNPs totally buried in glass with air as superstrate, which would result in a redshift of ~ 13 nm. However, in either of the single scattering scenarios with an air-glass interface such spectral shifts are anticipated, even when the interaction between the particle and its induced image is taken into account. Moreover, the particle-image interaction for the buried AuNPs is smaller than for the perfectly supported AuNPs due to the absence of a polarizable medium across the glass-substrate interface, thus only the latter is studied.

To verify whether the spectral position of the observed resonance in the experimental

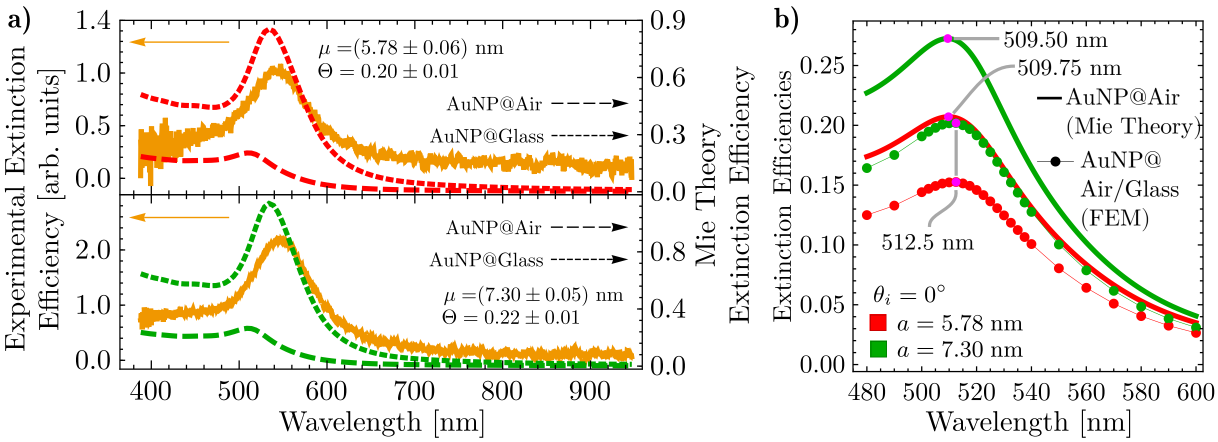

Fig. 2.2: **a)** Experimental extinction efficiency of the metasurfaces (continuous orange lines) when illuminated at normal incidence, and extinction efficiency of a single AuNP of radius $a = 5.78$ nm (red lines) and $a = 7.30$ nm (green lines) embedded in air (long dashed lines) and in glass (short dashed lines), as given by Mie Theory, as a function of the incident light wavelength. **b)** Extinction efficiencies as a function of the incidence wavelength, at normal incidence, of a single AuNP embedded in an infinite air matrix (continuous lines) calculated by means of Mie Theory, and in a semiinfinite air matrix supported on a semiinfinte glass substrate (markers) using FEM simulations. The red lines correspond to the calculations of a AuNP with radius $a = 5.78$ nm and the green lines to $a = 7.30$ nm. The magenta markers show the excitation of the LSPR for each case.

extinction spectra is influenced solely by the interaction between isolated AuNPs and the substrate, FEM calculations were performed. In Fig. 2.2b), extinction efficiencies (continuous lines) were calculated using standard Mie Theory for isolated AuNPs embedded in air considering radii of $a = 5.78$ nm (red) and $a = 7.30$ m (green) alongside the FEM simulations (red and green markers connected by thin lines), where the AuNPs were placed on a glass substrate ($n_s = 1.5$) and illuminated at normal incidence; magenta markers indicate the extinction efficiencies at the LSPR excitation wavelengths. It is observed that the LSPR appears at ~ 510 nm in the Mie Theory calculations for both radii, while the presence of the substrate causes a redshift of only ~ 3 nm. These results indicate that the resonances seen in the experimental extinction spectra of the metasurfaces in Fig. 2.2a) do not correspond to single-particle responses, but rather to collective excitations of the AuNPs forming the monolayer. To confirm that the spectral position of the observed resonances in the metasurfaces are a consequence of the multiple scattering among the AuNPs in the ensemble, measurements of the reflectivity of the metasurface under Attenuated Total Reflection (ATR) conditions were performed, so the AuNPs of the metasurface were illuminated with an evanescent wave enhancing its optical properties.

In Fig. 2.3, reflectivity spectra of the synthesized metasurfaces were measured in an ATR configuration as a function of incident wavelength. Measurements were conducted using s polarized (upper panels) and p polarized (lower panels) light at angles of incidence $\theta_i = 64.9°$ (chosen to be greater to the critical angle of a water-glass interface for future measurements) and $\theta_i = 72.9°$ (the maximum value achieved by the experimental setup). In addition, theoretical reflectivity curves (opaque lines) were calculated with the CSM with nominal parameters from Fig. 2.1. Moreover, a fitting of the CSM to the experimental data was performed to determine the best-fitting mean radius $a_{\mathrm{Fit}}^{\mathrm{CSM}}$ and best-fitting surface coverage $\Theta_{\mathrm{Fit}}^{\mathrm{CSM}}$ by minimizing the squared error with the Barzilai–Borwein gradient descent algorithm [29] under two constraints: a fixed surface coverage $\Theta_{\text{Fit}}^{\text{CSM}}$ for both polarizations per sample, and $\Theta_{\text{Fit}}^{\text{CSM}} 0 < 0.30$, to ensure validity of the CSM [23]. The resulting best-fitting parameters $a_{\text{Fit}}^{\text{CSM}}$ and $\Theta_{\text{Fit}}^{\text{CSM}}$ are shown in each panel of Fig. 2.3.

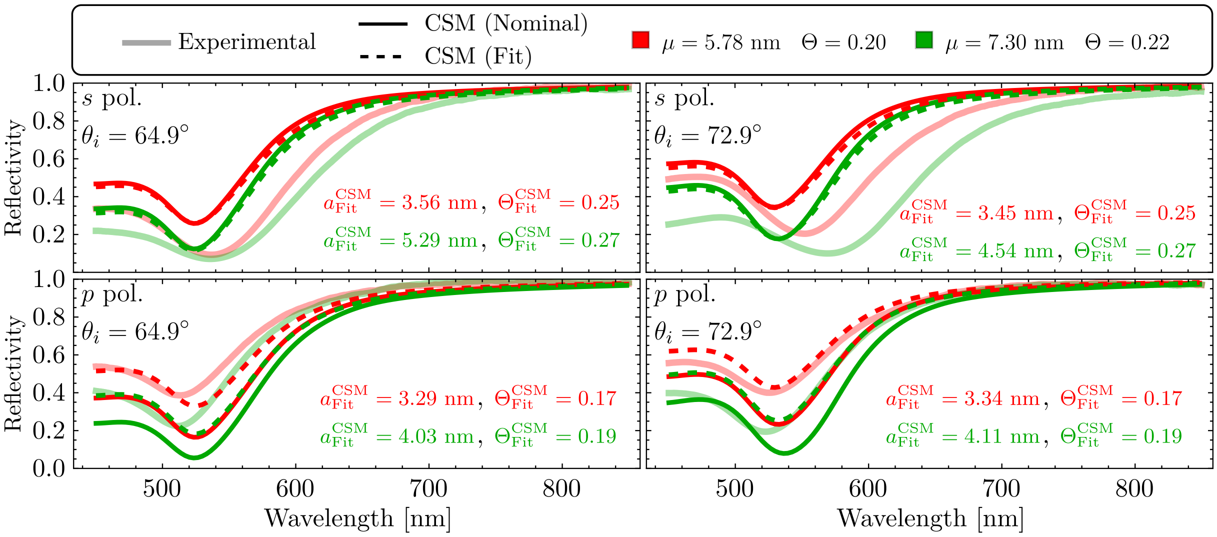

Fig. 2.3: Reflectivity at ATR configuration of a AuNPs metasurface as a function of the incident wavelength considering an angle of incidence $\theta_i = 64.9°$ (left column) and $\theta_i = 72.9°$ (right column), for s (upper row) and p (lower row) polarization state. The metasurface is embedded in a water matrix and supported onto a glass substrate with refractive indices $n_{\text{m}} = 1.33$ and $n_{\text{s}} = 1.5$, respectively. The experimental reflectivity (continuous translucent lines) of the metasurface is shown for the fabricated samples characterized by the mean radius μ of their size distribution and by its surface coverage Θ. The red lines correspond to the metasurface shown in Fig. 2.1a) while the green lines to the sample shown in Fig. 2.1b). The theoretical curves were calculated with the CSM employing the nominal values of the metasurface (opaque continuous lines), and employing the best-fitting parameters $a_{\text{Fit}}^{\text{CSM}}$ and $\Theta_{\text{Fit}}^{\text{CSM}}$ (dashed lines).

By comparing the experimental results with the calculations performed with the CSM employing the nominal values of the metasurfaces, a qualitative agreement related to surface coverage was observed: higher surface coverage resulted in greater light extinction in the visible reflection spectra. Also, a dependence of the spectral shift magnitude on the angle of incidence was identified, consistent with experimental results for p polarized light. Specifically, theoretical redshifts between samples were estimated to be approximately 3 nm and 4 nm for incident angles of $\theta_i = 64.9°$ and $\theta_i = 72.9°$, respectively, thus presenting a trend agreement for p polarization at oblique incidence. In contrast, no theoretical spectral shift between samples was observed for s polarized light at either angle, with all resonances appearing near 525 nm. Additionally, the overall reflectivity predicted by the CSM using nominal values was higher than the experimental measurements for s polarization, and lower for p polarization.

The values of the best-fitting parameters show the following trends for all considered cases. First, $a_{\text{Fit}}^{\text{CSM}}$ was found to be smaller than the nominal mean radius μ: For s polarization, $a_{\text{Fit}}^{\text{CSM}}$ increased with the angle of incidence, while for p polarization decreased. The fitted values satisfied the small particle approximation, consistent with the lack of spectral shift between samples observed in Fig. 2.3. The fitted surface coverage $\Theta_{\text{Fit}}^{\text{CSM}}$ also showed polarization dependence. Values greater than the nominal characterization were obtained for s polarization, while lower values were obtained for p polarization. However, all fitted parameters remained close to those obtained from morphological characterization, suggesting reasonable consistency between models and experimental data.

Reflectivities calculated using the best-fitting parameters (dashed lines in Fig. 2.3) showed a reduction in the s polarized case relative to predictions based on nominal values, though both exceeded the experimental data. For p polarization, the fitted model showed improved agreement with experiments across the visible spectrum, particularly for the sample with $\mu = 5.78$ nm at $\theta_i = 72.9°$. These results indicate that, while the CSM with fitted parameters improves the match with experimental reflectivity, it does not fully capture the optical response of the metasurfaces. Notably, polarization-dependent behavior observed in experiments is not reproduced by the CSM nor FEM simulations for a single particle on a substrate. This discrepancy suggests the presence of anisotropy in the system, potentially caused by partial embedding of the nanoparticles, an effect previously reported by Meng et al. [25] for thermally annealed AuNPs under similar conditions to the employed for the fabricated metasurfaces.

To evaluate the effect of a possible partial embedding of the metasurfaces, FEM simulations were performed for a single partially embedded AuNP with radius a, positioned at a height h and illuminated at normal incidence. The height h corresponds to the distance from the center of the AuNP to the glass-air interface, as shown in Fig. 2.4a), where r to represent the apparent radius of the AuNPs seen in SEM micrographs. To take into account a partial embedding of the AuNPs, it was considered an equivalent AuNP to be employed in the CSM with volume V_{eq} and

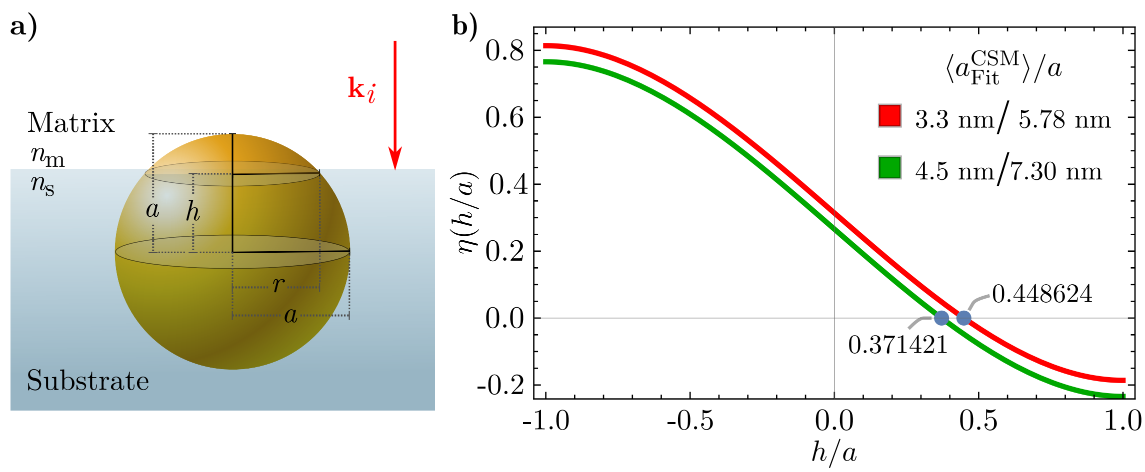

Fig. 2.4: **a)** Schematics of a AuNP of radius a partially embedded in a substrate, illuminated by an electromagnetic plane wave with an incident wave vector $\mathbf{k}_i$ perpendicular to the matrix-substrate interface. The distance between the center of the AuNP and the interface is defined as h while r is the aparent radius of the AuNP as seen from above. **b)** Plot of function $\eta(h/a)$ defined in Eq. (2.1) for a metasuface characterized for the most probable radius in its size distribution $a = 5.78$ nm and the average of the best fittig radius to the CSM $\langle a_{\text{Fit}}^{\text{CSM}} \rangle = 3.3$ nm (red line) and for a metasurfae with $a = 7.30$ nm and $\langle a_{\text{Fit}}^{\text{CSM}} \rangle = 4.5$ nm (green line). The roots of $\eta(h/a)$ are signalized with the blue markers for each case.

radius $a_{\mathrm{Fit}}^{\mathrm{CSM}}$, which are related to the fraction of the AuNPs above the substrate $V_{\mathrm{above}}^{\mathrm{semi\text{-}sphere}}$ as

$$V_{\mathrm{above}}^{\mathrm{semi\text{-}sphere}} = \frac{4}{3}\pi a^3 \left[\frac{1}{2} - \frac{3}{4}\left(\frac{h}{a}\right) + \frac{1}{4}\left(\frac{h}{a}\right)^3\right] = \frac{4}{3}\pi (a_{\mathrm{Fit}}^{\mathrm{CSM}})^3 = V_{\mathrm{eq}}, \tag{2.1}$$

where $h = a$ corresponds to a sphere totally embedded in the substrate and $h = -a$ to one perfectly supported on it. The partial embedding of the metasurface can be calculated, in average, by solving for h in Eq. (2.1), which is equivalent to calculate the roots of the function η, given by

$$\eta\left(\frac{h}{a}\right) = \frac{1}{2} - \frac{3}{4}\left(\frac{h}{a}\right) + \frac{1}{4}\left(\frac{h}{a}\right)^3 - \left(\frac{a_{\mathrm{Fit}}^{\mathrm{CSM}}}{a}\right)^3 = 0. \tag{2.2}$$

In particular, in Fig. 2.4b) the function $\eta(h/a)$ is plotted considering the average of $a_{\mathrm{Fit}}^{\mathrm{CSM}}$ for each metasurface characterized by the most probable radius a of their AuNPs. The respective values of such parameters are presented in Fig. 2.4b) and its roots are indicated with the blue markers. Since in both considered cases the roots are positive, it can be concluded that most of the volume of the AuNPs of the metasurface are, in average, inside the substrate.

Fig. 2.5 presents the extinction efficiencies of single AuNPs with radii $a = 5.78$ nm [Fig. 2.5a)] and $a = 7.30$ nm [Fig. 2.5b], partially embedded in a glass substrate and surrounded by air, as a function of the incident wavelength under normal illumination. The degree of embedding is defined by the ratio h/a, with representative values taken from Fig. 2.4b). Three embedding scenarios were analyzed: predominantly in air ($h/a < 0$, circles), mostly in glass ($h/a > 0$, diamonds), and symmetrically embedded ($h/a = 0$, squares). For comparison, the extinction spectra of the corresponding isolated AuNP in air, calculated with standard Mie theory, are

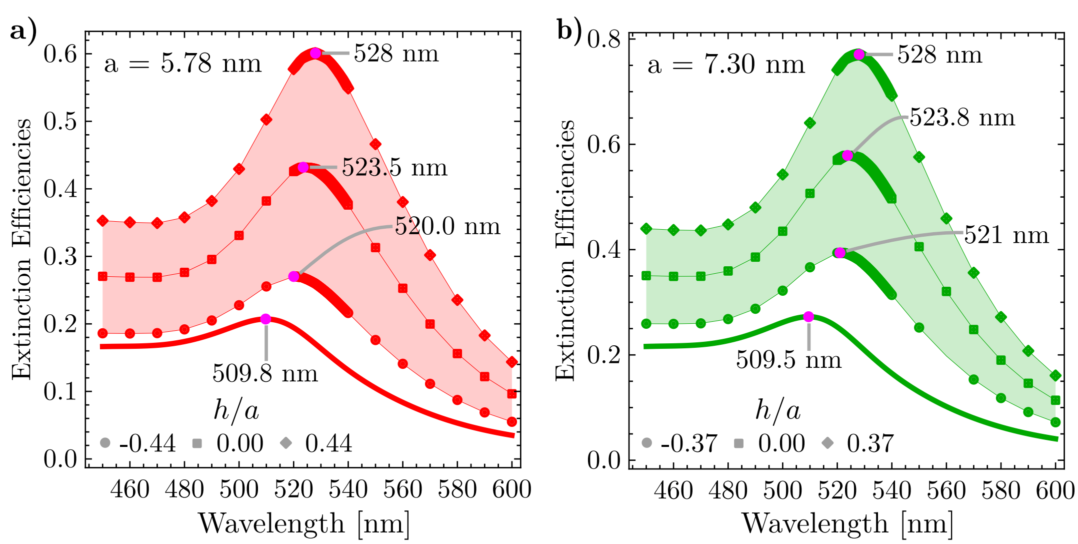

Fig. 2.5: Extinction efficiencies as a function of the incident wavelength of a single AuNP, partially embedded in a glass substrate and an air matrix, of radius **a)** $a = 5.78$ (red lines) nm and **b)** $a = 7.30$ nm (green lines) when illuminated at normal incidence relative to the air-glass interface. The partial embedding of the AuNPs is characterized by the quantity h/a whose nominal value is specified for each AuNP according to Fig. 2.4. Three values of partial embedding where considered: AuNP is mostly embedded in air (disks), in glass (diamonds) and equally embedded in both media (squares).

shown as solid lines. The magenta markers correspond to the spectral position of th LSPR for each considered case.

By comparing the resonance wavelength of the isolated AuNP with the partially embedded AuNP, a redshift between ~ 8 nm and ~ 20 nm is observed for both particle sizes and all embedding degrees. The redshift caused by the partial embedding is thus larger than the redshift calculated between the isolated and the supported AuNP on the substrate in Fig. 2.2b). However, the sole effect of the partial embedding does not reproduce the discrepancies in the extinction spectra in Fig. 2.2a) of ~ 36 nm since the contribution of the multiple scattering is yet to be considered. As a first numerical approach to study multiple scattering effects in a partially embedded metasurface, FEM calculations of a AuNP dimer partially embedded were performed.

The considered partially embedded dimer consisted of two identical AuNPs with radii $a = 7.30$ nm equally embedded in an air matrix and glass substrate ($h/a = 0$) and separated by varying interparticle gaps. The dimer was illuminated by an electromagnetic plane wave traveling normally into the glass-air interface, with its polarization aligned along the dimer's symmetry axis. In Figs. 2.6a) and 2.6b) the magnitude of the total electric field $\|\mathbf{E}_{\mathrm{Tot}}\|$ was calculated for the dimmer considering two gap distances of 4 nm and 14 nm, respectively. Additionally, in Fig. 2.6c) the extinction cross section of the dimmer for various interparticle gaps as a function of the incident wavelength are presented; the cross section of a single AuNP (red continuous line), calculated using Mie theory, is shown as reference. The magnitude of the total electric field in Figs. 2.6a) and 2.6b) is evaluated at the wavelength of the LSPR for each case, corresponding to the wavelength of the maxima in the extinction spectra.

From Figs. 2.6a) and 2.6b) it can be seen that the electric field enhancement is greater in the glass substrate than in the air matrix. This enhancement is attributed to the increased apparent area of the AuNPs within the substrate due to partial embedding into the optically denser medium. When extended to a metasurface composed of many such particles, this effect implies that the effective surface coverage is larger than what would be expected from geometry alone. Also, this effect is expected to be stronger when the incident electromagnetic plane wave is s polarized than p polarized, as suggested by the best-fitting parameters in the CSM since the electric field has only in-plane components relative to the metasurface. When considering the spectral behaviour of the partially embedded dimer in Fig. 2.6c), it can be seen that the redshift relative to the isolated AuNP equals ~ 20 nm for the farthest separated dimer with a 14 nm gap (purple line), while for the closest dimer it grows up to ~ 30 nm (red line). Therefore, when considering the contribution from the partial embedding and the multiple scattering together, spectral shifts at normal incidence can be expected to behave as observed in the experimental extinction spectra presented in Fig. 2.2a).

The FEM simulations confirmed that the redshift observed in the experiments can not be attributed solely to substrate interactions in the single-particle case, emphasizing the necessity of incorporating multiple scattering effects. Fittings to the experimental data with the Coherent Scattering Model (CSM) revealed an apparent polarization-dependent variation in nanoparticle size and an overall increase in surface coverage, improving agreement with experimental data. However, discrepancies between the resonance wavelengths predicted by the CSM and those measured experimentally persist, likely due to structural features not resolved in the available SEM images. Based on experimental observations, FEM simulations, and CSM predictions, it is suggested that the gold nanoparticles are partially embedded in the substrate. This partial

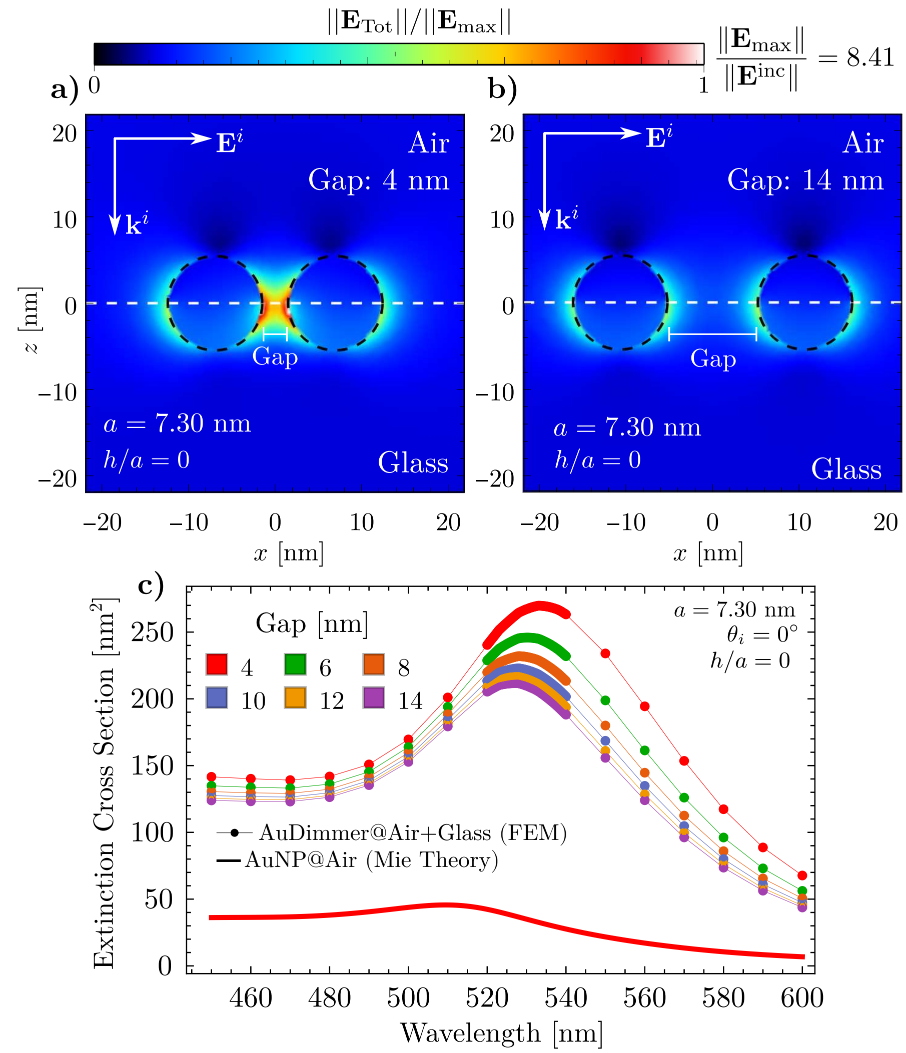

Fig. 2.6: Magnitude of the total electric field $\|\mathbf{E}_{\mathrm{Tot}}\|$ of a AuNP dimmer of radius $a = 7.30$ nm separated by a gap between particles of **a)** 4 nm and **b)** 14 nm, illuminated at normal incidence by an electromagnetic plane wave polarized along the symmetry axis of the dimmer. The AuNPs are partially embedded equally in an air matrix and a glass substrate ($h/a = 0$). **c)** Extinction cross section of the equally partially embedded AuNP dimer of radius $a = 7.30$ nm as a function of the incident wavelength; the color of each curve corresponds to different values of the distance between the AuNPs of the dimmer.

embedding alters the apparent particle size and increases the effective surface coverage, as estimated through FEM calculations for individual particles. A comprehensive theoretical model that incorporates nanoparticle embedding into frameworks like the CSM has yet to be developed, and new approaches are still required.

2.2 Partial Embedding: First Measurements

In the previous Section, the optical response of AuNP metasurfaces was analyzed taking into account substrate effects under realistic conditions. A partial embedding, enhanced by multiple scattering within the AuNPs ensemble, was proposed as the primary mechanism responsible for the spectral shifts observed in experimental measurements relative to theoretical predictions based on idealized scenarios. To further explore this hypothesis, two sets of metasurfaces were fabricated by collaborators at INAOE as explained at the beginning of Section 2.1 considering two different annealing temperatures for each set. A controlled average partial embedding of the metasurface can be achieved with different annealing temperature due to the melting point of the used substrate. In this Section, the first optical characterization[1] of the fabricated samples are presented, however it is emphasized that SEM and AFM images are still required to determine the size distribution of the AuNPs and to estimate the partial embedding of the metasarfaces.

In Fig. 2.7 it is shown the transmissivity spectra at normal incidence ($\theta_i = 0°$) of the synthesized AuNPs metasurfaces fabricated by thermally annealing continuous gold films on Corning glass of different nominal thicknesses d. The thermal treatment was performed at two temperatures —500° [Fig. 2.7a)] and 550° [Fig. 2.7b)]— on gold films varying from 3 nm of thickness to 11 nm in steps of 2 nm based on the work of Meng et al. [25]. The color of each curve corresponds to one value of the initial thicknesses of the samples.

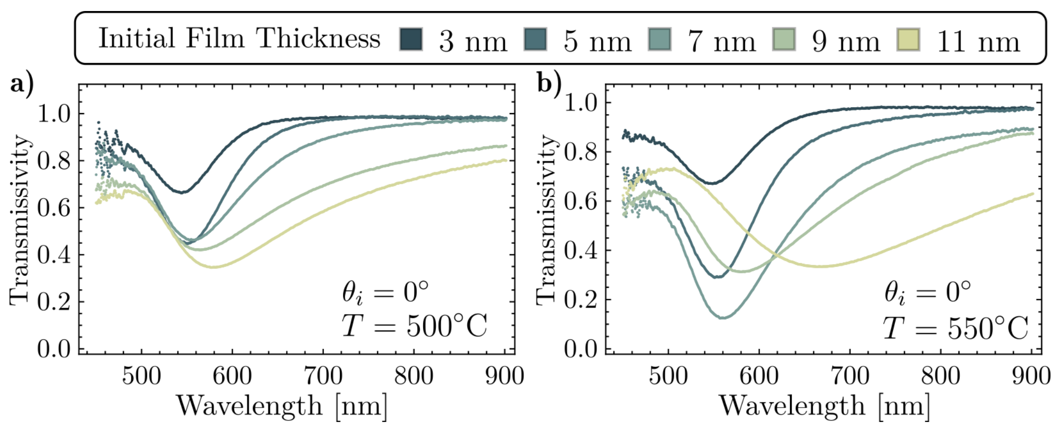

Fig. 2.7: Transmissivity spectra of the synthesized metasurfaces after thermal annealing at a temperature of **a)** 500° and **b)** 550° at normal incidence ($\theta_i = 0°$) as a function of the incidence wavelength. The color legend on top indicates the nominal thicknesses of the initial deposited gold films prior the dewetting process.

For either of the analyzed sample sets, that is, considering a fixed annealing temperature, the extinction spectra minima exhibit a redshift that increases as the initial thickness of the film does for all considered samples. On the other hand, only for the metasurface synthesized at 500°C it is observed that the initially thicker the samples, the lower their transmissivity within the visible spectrum, while for the 550°C synthesized samples, this trend is followed up to the 7 nm thick sample. This results are consistent with the fabricated samples addressed in Section

[1] It is stressed that the measurements presented in Section 2.1 were performed by collaborators at INAOE, while I conducted the measurements presented in Section 2.2.

2.1 where SEM images were available, thus larger AuNPs are expected in the metasurface for thicker films. It is noteworthy that for all the synthesized samples at 500°C, the transmissivity minima are spectrally located in the visible range at wavelengths $\lambda < 600$ nm while for the samples annealed at 550°, this is not observed for all samples. The metasurface fabricated with an initial thickness of $d = 11$ nm —yellowish line in Fig. 2.7b)— present the broadest and most redshifted minimum of all samples at ~ 680 nm, which suggest the synthesis of non-spherical particles [9] or particles with a greater size in less dense ensembles [42].

When taking into account the temperature dependency on the dewetting process, it can be seen that the spectral behavior of the samples become more pronounced for the highest annealing temperature. For example, the transmissivity evaluated at the mininma in Fig. 2.7b) is smaller than in Fig. 2.7a), that is the extinction is maximized for the metasurfaces fabricated at 550° than at 500°. Additionally, for the same initial thickness value, the observed minima for the samples heated at 550°C are redshifted compared to those identified for the samples heated at 500°C. Both of these behaviors can be explained by a partial embedding as discussed in Section 2.1 under the assumption that resulting AuNPs synthesize from a fixed d are of similar size and that the partial embedding can be achieved by a softening process of the Corning glass substrate for higher temperatures. To further support the hypothesis of the partial embedding based in the theoretical work presented in Section 2.1, the reflection spectra in an Attenuated Total Reflection (ATR) configuration of the synthesized metasurfaces were measured.

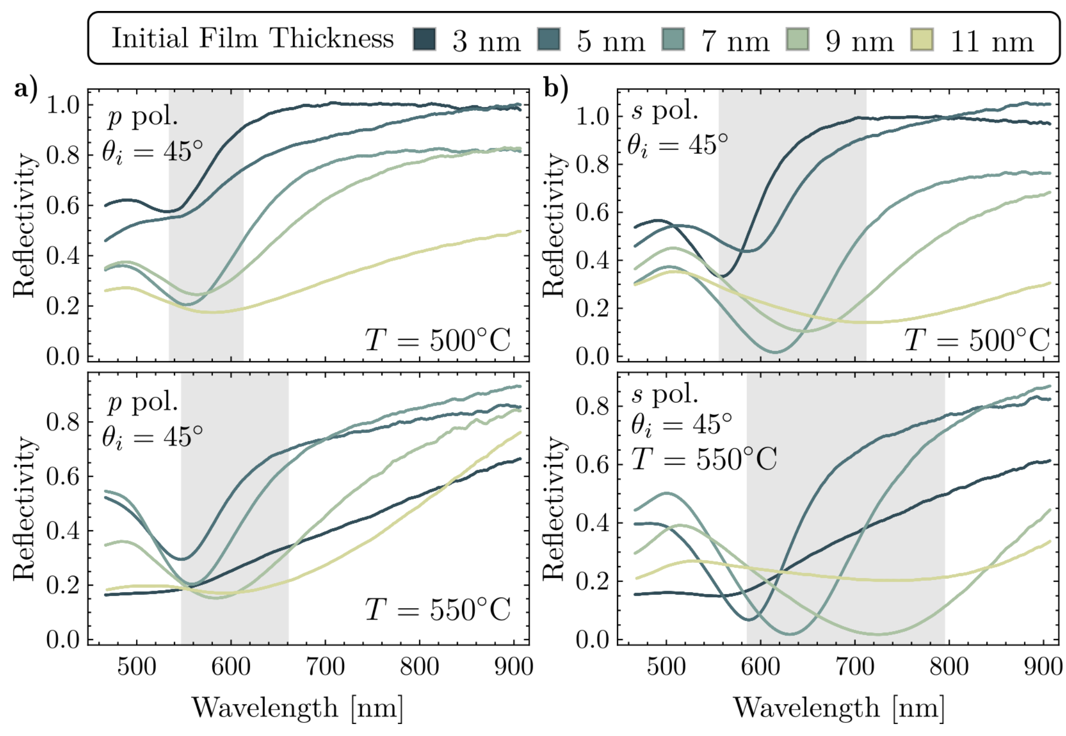

Fig. 2.8: Reflectivity spectra in an ATR configuration for **a)** p polarized and **b)** s polarized light as a function of the incident wavelength at an angle of incidence $\theta_i = 45°$ for the synthesized AuNPs metasurfaces. The analyzed samples were fabricated by thermally annealing gold thin films with varying initial thicknesses d (color code) at a temperature of 500°C (top panels) and of 550°C (bottom panels). The shaded areas correspond to the spectral windows where the reflectivity minima are located.

The reflectivity spectra of the fabricated metasurfaces in an ATR configuration is presented in Fig. 2.8 for an angle of incidence $\theta_i = 45°$. The measurements were performed with incident light in a p polarization state [see Fig. 2.8a)] and in an s polarization state [see Fig. 2.8b)]. The top panels correspond to measurements for the thermally annealed metasurfaces at 500°C and the bottom panels to the 550°C. The shaded regions in all panels highlight the spectral window in which the minima of the reflectivity are located.

For each independent panel shown in Fig. 2.8, it can be seen that the reflectivity minima for 5 nm $\leq d \leq$ 11 nm is redshifted as the initial thickness of the film increases, in accordance to the previous results for the extinction spectra. When the position of these minima are compared for the same samples but different polarization states, that is by comparing the panels in the same row of Figs. 2.8a) and 2.8b), it follows that the observed redshifts are larger for the s polarization case, which can be easily visualized by noticing the sizes of the shaded regions. Moreover, for each polarization state the observed minima are farther redshifted for the thermally annealed metasurface at 550°C (bottom panels) than those heated at 500°C (top panels). In the described spectral behavior of the metasurfaces's redshifts, the main effects due to a partial embedding discussed in Section 2.1 can be identified, specifically for the samples heated at the greater temperature, which is expected to present a higher embedding degree.

As stated a the beginning of this Section, the presented results are the first measurements performed aimed to characterize the partial embedding of the metasurface. While the obtained experimental spectra are consistent with the theoretical and numerical results from Section 2.1, a morphological characterization of the fabricated samples are needed in order to follow a similar methodology. To further explore the optical response of a metasurface presenting partial embedding from an experimental approach, the synthesis of more samples with varying annealing temperatures are required accompanied by cross-sectional SEM and AFM images. From a theoretical standpoint, it is proposed to incorporate the partial embedding into analytical frameworks —such as the Coherent Scattering Model or the Thin Film Theory— for the surrounding the AuNPs aided with numerical simulations for single scatterers partially embedded under polarized illumination.

Chapter 3

Progress in Optical Response and Sensing Mechanisms of Metasurfaces

This Chapter presents advances in the study of the optical response of plasmonic metasurfaces composed of disordered ensembles of spherical gold nanoparticles (AuNPs), as well as the physical mechanisms that enable these systems to be used in biosensing applications. The results consist in theoretical calculations performed with the Coherent Scattering Model (CSM) and the Thin Film Theory (TFT), and experimental measurements —which I conducted during a short fellowship at the Instituto Nacional de Astrofísica, Óptica y Electrónica (INAOE)—. The theoretical results were aimed at the design of metasurfaces based on a targeted spectral response, while the obtained experimental data consists in reflectivity spectra supporting the presence of a resonance identified in the theoretical findings. In Section 3.1, two metasurface designs are proposed, along with their expected spectral behavior as predicted by the CSM and the TFT. Additionally, the use of ellipsometric parameters —instead of conventional reflectivity spectra— is discussed, motivated by the phenomenon of Topological Darkness [30] and the ease of performing such measurements in possible high-sensitivity detection systems [7]. Lastly, the expected spectral shift in the optical response of the metasurface due to variations on a probe medium were determined, which predict a blueshift of the identified resonance of the metasurface. To verify this phenomenon experimentally, in Section 3.2 the reflectivity spectra was measured for a metasurface exposed to a medium, whose refractive index is artificially varied.

3.1 Design of Metasurfaces and Use of Topological Darkness for Biosensing

The choice of a plasmonic metasurface of disordered spherical AuNPs for biosensing was motivated by its cost-effectiveness and rapid fabrication process compared to ordered nanoparticle arrays or AuNPs with more complex geometries [5, 7]. To design the biosensing-aimed metasurface with the desired morphological characteristics, the mean radius a of the AuNPs and the surface coverage Θ of the metasurface have to be determined. These parameters must ensure that the optical response of the metasurface lies within the visible spectrum, as biological elements can be damaged by ultraviolet light and are typically immersed in aqueous media [5], which exhibit significant absorption in the infrared region.

Two different designs for AuNPs disordered metasurfaces are proposed based on the available fabrication and measurement capabilities at INAOE, and on exhibiting a spectral response within the visible range. The values of a and Θ for the designs were determined through a parametric variation procedure using reflectivity spectra calculations performed with the CSM and the TFT formalisms, each exploiting their modeling strengths. For the CSM-based design, the values $a = 30$ nm and $\Theta = 0.12$ were chosen, corresponding to AuNPs beyond the dipolar approximation in relatively sparse ensembles. In contrast, the TFT-based design resulted in $a = 17.5$ nm and $\Theta = 0.275$, suggesting smaller AuNPs forming denser ensembles.

In Fig. 3.1a) the reflectivity at an angle of incidence $\theta_i = 72°$ of the proposed metasurfaces is presented as a function of the incident wavelength λ within the visible spectra for p (continuous lines) and s polarized (dashed lines) light. The upper and lower panels in Fig. 3.1 correspond to calculations with the CSM and the TFT with their corresponding parameters for the proposed metasurfaces, respectively, employing a substrate with a refractive index $n_\mathrm{s} = 1.5$. Each curve in Fig. 3.1 corresponds to a different refractive index n_m of the matrix, as specified by the color scale in Refractive Index Units (RIU), considering water as the reference matrix with $n_\mathrm{m} = 1.33$ and changing up to $n_\mathrm{m} = 1.36$. The shaded regions in Fig. 3.1a) highlights the spectral window of 500 nm $\leq \lambda \leq$ 590 nm and their manifications are shown in Fig. 3.1b), enabling a better visualization of spectral shifts induced by variations in the matrix refractive index.

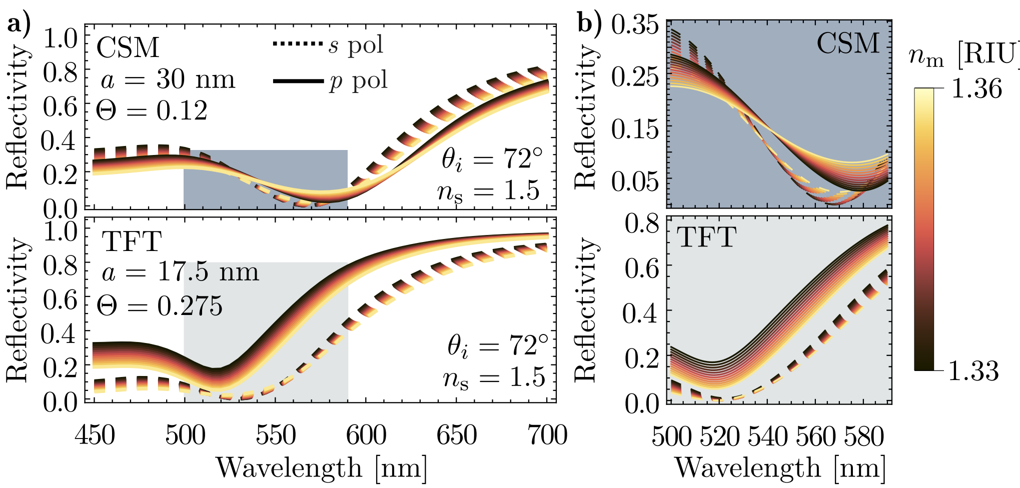

Fig. 3.1: Theoretical reflectivity spectra as a function of the incidence wavelength **a)** in the visible spectrum and **b)** in the spectral window between 500 nm and 590 nm of a metasurface characterized by a mean radius $a = 30$ nm and surface coverage $\Theta = 0.12$ (upper panels, calculated with the CSM) and one metasurface characterized by $a = 17.5$ nm and $\Theta = 0.275$, (lower panels, calculated with the TFT). A glass substrate with refractive index $n_\mathrm{s} = 1.5$ and an angle of incidence $\theta_i = 72°$ were considered. The continuous (dashed) lines correspond to p (s) polarized light and each curve corresponds to varying matrix refractive indices n_m, as indicated by the color legend. The shaded region highlights the window between 500 nm and 590 nm.

From its theoretical design, the proposed metasurfaces fulfill the spectral behavior requirements for biosensing applications. On the one hand, the metasurface designed under the CSM conditions (upper panels) exhibits a resonances at wavelengths near 580 nm for p polarization and near 570 nm for s polarization for all considered values of n_m. On the other, the metasurfaces proposed under the TFT features present the resonances at approximately 520 nm and 525 nm for p and s polarization, respectively. For both metasurfaces and polarizations, a track of the minima spectral shift can be done, enabling the implementation of an intensity-based sensing scheme. However, the broad resonance widths observed in Fig. 3.1b) reduce the quality factor of the simulated sensing mechanism, potentially limiting the sensitivity of the measurements. As an alternative, a phase-based sensing scheme based on the Topological Darkness (TD) phenomenon and on measurements of ellipsometric parameters offers improved quality factors [7].

The TD phenomenon refers to the occurrence of zero-reflection at specific spectral points in metasurfaces, arising from destructive interference or mode coupling induced by their nanostructured elements [7, 30]. The term TD reflects the robustness of the light suppression condition against perturbations, due to topological constraints present in the reciprocal space of the metasurface's optical response [30], thus making metasurfaces exhibiting TD highly attractive for sensing applications [7]. While the TD has been primarily studied in ordered systems identifying the zero-reflection, it has also been demonstrated in disordered metasurfaces considering a near zero-reflection condition instead [9, 13]. Moreover, the TD phenomenon is associated with abrupt phase changes in the reflected light spectra at the zero-reflection wavelength, which shifts with changes in the refractive index of the matrix [7, 9, 30], as seen in Fig. 3.1b).

Fig. 3.2a) presents the phase difference $\Delta\phi$ as a function of the incident wavelength and the angle of incidence θ_i for the proposed metasurfaces illustrating the TD phenomenon. The left (right) panel displays the calculated reciprocal space of the CSM-based (TFT-based) design considering a water matrix ($n_\mathrm{m} = 1.33$). Spectral points exhibiting TD are signalized by the white loops with defined orientation. The white continuous and dashed lines indicate angular cuts at $\theta_i = 72°$ corresponding to the phase difference curves shown in Fig. 3.2b) using the same line styles.

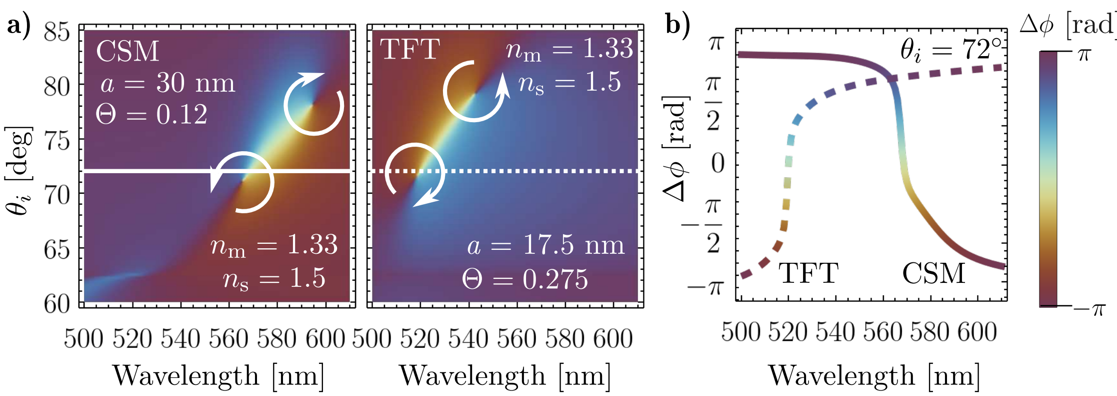

Fig. 3.2: Ellipsometric parameter $\Delta\phi$ **a)** as a function of the incident wavelength and the angle of incidence θ_i and **b)** as a function of the incident wavelength at $\theta_i = 72°$ for AuNPs metasurfaces suported on a glass ($n_\mathrm{s} = 1.5$) and embedded in water ($n_\mathrm{m} = 1.33$), whose designed was based on the conditions of the CSM (left panel, $a = 30$ nm and $\Theta = 0.12$) and of the TFT (right panel, $a = 17.5$ nm and $\Theta = 0.275$). The white arrows indicate spectral points where $\Delta\phi$ is not well-defined and their flow show the increase in the phase difference. The continuous and dashed white lines correspond to the spectral cuts of $\Delta\phi$ at $\theta_i = 72°$ for the CSM and the TFT calculations, respectively.

The TD phenomenon can be identified for each panel in Fig. 3.2a) based on the following features. First, the near zero-reflectance condition is fulfilled at two spectral points [see white loops in Fig. 3.2a)] for each mestasurface design, as indicated by the reflectivity spectra at $\theta_i = 72°$ in Fig. 3.1, in consistency with Eq. (1.34) when $r^{(p)} \to 0$. In addition, $\Delta\phi$ at these spectral points becomes not well-defined, exhibiting an abrupt phase change as observed in the angular cuts shown in Fig. 3.2b). The third TD feature observed in the theoretical results is the topological charge of ±1 associated with each loop in the phase space of the metasurfaces [30], as observed in each panel of Fig. 3.2a). The vanishing net topological charge ensures that any angular cut crossing the bright region between the TD points in Fig. 3.2b) will display an inflection point in the phase difference, as exemplified in Fig. 3.2b). This property enables a robust and stable phase-based sensing mechanism.

In Fig. 3.3 it is presented the phase-based sensing scheme with the proposed metasurfaces for a varying refractive index n_{m} of the matrix at two angles of incidence: 70° (blue gradient) and 72° (red gradient). The upper panels correspond to the CSM-based design and the lower ones to the TFT-based design. Fig. 3.3a) shows the phase difference $\Delta\phi$ as a function of the incident wavelength λ where each curve corresponds to different values of n_{m}. In Fig. 3.3b) the spectral derivative of this ellipsometric parameter is calculated, that is, $\mathrm{d}\Delta\phi/\mathrm{d}\lambda$, whose maxima indicate the occurrence of TD at a specific wavelength λ^{TD}. The spectral shift $\delta\lambda^{\mathrm{TD}}$

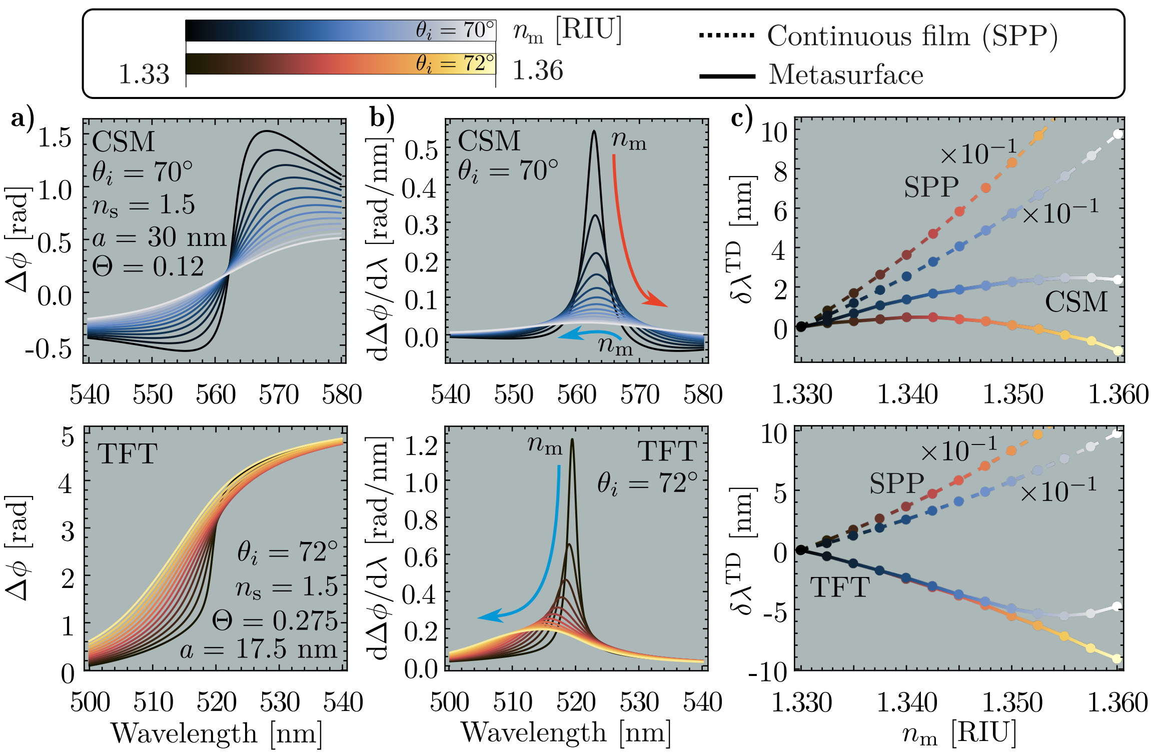

Fig. 3.3: Phase-based sensing scheme with a AuNPs metasurface consisting in **a)** the phase difference $\Delta\phi$ spectrum and **b)** tis spectral derivative $\mathrm{d}\Delta\phi/\mathrm{d}\lambda$, both as a function of the incident wavelength for changes in the refractive index n_{m} of the matrix at an angle of incidence θ_i of 70° (blue gradient) and 72° (red gradient). The upper panel shows the results with the CSM ($a = 30$ nm, $\Theta = 0.12$ and $\theta_i = 70°$) and the lower panel with TFT ($a = 17.5$ nm, $\Theta = 0.275$ and $\theta_i = 72°$). The maxima in the $\mathrm{d}\Delta\phi/\mathrm{d}\lambda$ spectra are located at the TD occurrence wavelength λ^{TD} and their tracking is signalized by the red and blue arrows. **c)** Spectral shift $\delta\lambda^{\mathrm{TD}}$ as a function of n_{m} for the proposed matasurfaces (continuous lines) at $\theta_i = 70°$ and $\theta_i = 72°$; the spectral shift of the SPP, scaled by a factor of 10^{-1}, excited on a 50 nm thick continuous gold film (dashed lines) is shown for comparison under an abuse of notation.

in response to changes in n_{m} is indicated by arrows. Lastly, in Fig. 3.3c) the spectral shift $\delta\lambda^{\mathrm{TD}}$ of the metasurfaces (continuous lines) is shown as a function of n_{m} for the two selected angles of incidence with the different color gradients. For comparison, the theoretical spectral shift of the Surface Plasmon Polariton (SPP) excited on a continuous gold film, considering an optimal thickness of 50 nm [43], is shown in dashed lines[1] in Fig. 3.3c). The presented SPP spectral shift is scaled by a factor of 10^{-1}, indicating higher sensitivity compared to the metasurface. However, it has been reported that plasmonic metasurfaces can outperform thin films in sensing performance, particularly at low concentrations of the target substance, due to the light confinement on the surface of the nanostructures conforming the metasurfaces [44].

From the results shown in Fig. 3.3a) the robustness of the phase-based sensing can be identified by comparing the colored curves from darkest to lightest at $\lambda^{\mathrm{TD}} \sim 560$ nm and at ~ 520 nm for the CSM and TFT calculations, respectively. The phase changes evaluated at λ^{TD} become less pronounced as n_{m} increases for a fixed θ_i, however a change in concavity is still observed. These concavity changes are more easily identified in the $\mathrm{d}\Delta\phi/\mathrm{d}\lambda$ spectra, since their maxima are located at λ^{TD}, where the change of concavity in the $\Delta\phi$ plots occur. Thus, phase-based sensing curves can be obtained by tracking λ^{TD} for different values of n_{m} with maxima-finding algorithms applied to the $\mathrm{d}\Delta\phi/\mathrm{d}\lambda$ spectra. Additionally, the quality factors of the spectral response in the $\mathrm{d}\Delta\phi/\mathrm{d}\lambda$ curves at λ^{TD} overcomes their intensity-based counterparts in Fig. 3.1b) due to narrower peaks, particularly for small changes in n_{m} (darker curves). The tracking of λ^{TD} in the $\mathrm{d}\Delta\phi/\mathrm{d}\lambda$ spectra—indicated by the color and direction of the inset arrows shown in Fig. 3.3b)— reveals both a red- and a blueshift for the CSM calculation and only a blueshift in the TFT results. As shown in Fig. 3.3c), the spectral shift $\delta\lambda^{\mathrm{TD}}$ at the two angles of incidence —70° (blue gradient) and 72° (red gradient)— confirms that both red- and blueshifts can occur in either of the proposed metasurface by selecting the angle of incidence, in contrast with the SPP case, where only redshifts are observed.

In this Section, the design for two AuNPs metasurfaces was proposed based on two different morphological criteria: large AuNPs in sparse ensembles and small AuNPs in dense ensembles. The design parameters were optimized through parametric simulations using the CSM and TFT frameworks, respectively. The advantages of a phase-based sensing scheme over an intensity-based scheme were discussed, and the former was simulated for both proposed metasurfaces taking advantage of the TD phenomenon. The theoretical response of the metasurfaces predicted a spectral blueshift for an increasing matrix refractive index, contrasting with the well-established redshift of the SPP. While these findings highlight the potential of disordered metasurfaces for phase-based sensing, the physics behind the observed blueshift remains an open question. To experimentally validate this spectral behavior, intensity-based measurements for an AuNPs metasurface are presented in the following Section.

[1]The inclusion of the SPP spectral shift serves only as a reference, since the resonance wavelength of the SPP does not originate from the TD phenomenon and it is determined with an intensity-based sensing scheme. Therefore, the symbol $\delta\lambda^{\mathrm{TD}}$ for the SPP spectral shift becomes an abuse of notation.

3.2 Experimental Observation of Spectral Blueshift

In the previous Section, a blueshift in the optical response of a disordered AuNPs metasurfaces was theoretically predicted for an increase in the refractive index n_{m} of the matrix surrounding the AuNPs. The calculations were performed using Multiple Scattering Theories (MSTs) such as the Coherent Scattering Model and the Thin Film Theory, both of which have previously shown qualitative agreement with experimental results. However, no experimental evidence of a blueshift has been reported. To validate the predicted blueshift with the employed MSTs, the reflectivity spectra of a metasurface —yet to be morphologically characterized— were performed for a varying matrix.

In Fig. 3.4 it is shown the reflectivity spectra in an Attenuated Total Reflection (ATR) configuration of the metasurface sample under p [see Fig. 3.4a)] and s polarized illumination [see Fig. 3.4b)] at an angle of incidence $\theta_i = 64.9°$, which is larger than critical angle for a water-glass interface. Each curve in the main panels correspond to measurements performed with a matrix of aqueous solutions of sodium chloride at concentrations between 0% and 11%; their corresponding values of n_{m} were determined with an Abbe refractometer (WY1A MG Scientific). The shaded regions in the main frames highlight the spectral windows where the reflectivity minima are located and their magnification is presented in the lower left panels for each polarization. The lower right panels display the resonance wavelength shift $\delta\lambda^{\mathrm{res}}$ as a function of n_{m}, where the blueshift of the resonance for the p polarization case can be observed.

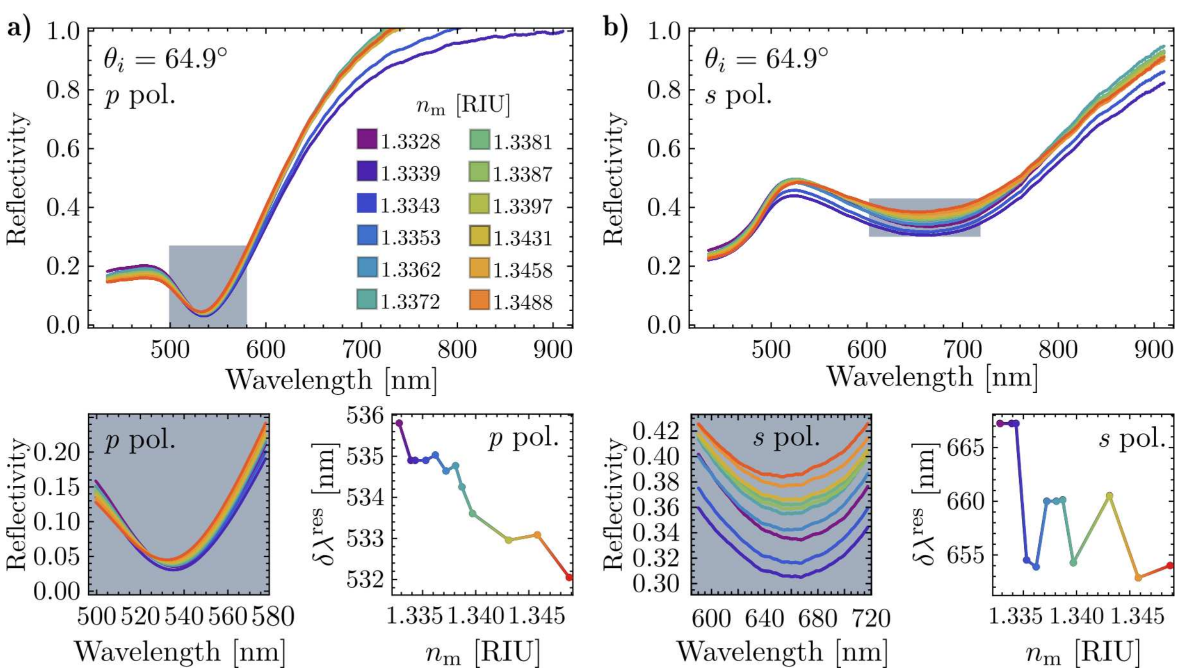

Fig. 3.4: Experimental reflectivity spectra in an ATR configuration as a function of the incident wavelength for **a)** p **b)** s polarized light at an angle of incidence $\theta_i = 64.9°$ of a AuNP metasurface. Each curve in the main frames correspond to measurments with different values of the matrix refractive index n_{m}. The shaded regions in the main frames highlight spectral windows were the minima of the reflectivity are found are their magnification is shown in the lower left panels for each polarization. The lower right panels show the resonance wavelength shift $\delta\lambda^{\mathrm{res}}$ as a function of n_{m}.

The experimental data in Fig. 3.4 confirms the presence of a blueshift of the observed resonances at the resonance wavelength λ^{res} with increasing matrix refractive index for a disordered AuNP metasurface. While a characterization of the average radius of the AuNPs and the surface coverage is required for a more comprehensive analysis, these results support the theoretical observations in the past Section. Additionally, measurements of ellipsometric parameters are encouraging to investigate the role of the Topological Darkness in the blueshift, that is, to evaluate the interference as a trigger of this spectral shift. Lastly, it is proposed to estimate if Effective Medium Theories (EMTs) modeling disordered AuNPs metasurfaces are able to reproduce the observed blueshift as achieved by the employed MSTs, highlighting the physical mechanisms enabling the predicted and observed spectral blueshift.

To continue the work presented in this Chapter, two main courses of action are being pursued. On the experimental side, a short fellowship at INAOE its being arranged, with the goal to conduct phase-change measurements of light reflected by AuNPs metasurfaces, followed by phase-based sensitivity tests. Additionally, I am beginning to get involved in the fabrication processes of AuNPs metasurfaces through dewetting and through chemical synthesis methods, to corroborate the viability of the proposed metasurfaces presented in Section 3.1. On the theoretical side, it is being explored if the radiative corrections within the dipolar approximation [46, 47] to the AuNPs in the metasurface can explain the observed blueshift predicted by the employed MSTs in Section 3.1 and experimentally observed in Section 3.2.

Chapter 4

Published and In-preparation Articles

During my doctoral studies, I contributed to the publication of the following peer-reviewed work:

Phase Singularity in Random Gold Metasurface for Refractive Index Sensing: Experimental Demonstration and Theoretical Analysis [9]

J. P. Cuanalo-Fernández, N. Korneev, I. Cosme, M. B. De La Mora, **J. A. Urrutia-Anguiano**, A. Reyes-Coronado, R. Ramos-García, and S. Mansurova

J. Phys. Chem. C **128**(27): 11372–11381, 2024.

DOI: 10.1021/acs.jpcc.4c02274

The phenomenon of topological darkness has been experimentally observed and theoretically characterized in metasurfaces with ordered spatial distributions of their building entities and exploited as a high sensitive sensing mechanism. In this article, the conditions to obtain topological darkness —near zero reflectance and phase singularity— are fulfilled in a random array of gold nanoislands (oblate ellipsoids) using theoretical and experimental methods. Reflectance and phase spectra were measured using a common-path interferometer in an attenuated total reflectance setup. Results reveal a partial state of topological darkness, with phase singularities marked by abrupt $\pm\pi$ phase jumps. Additionally, the study highlights the potential of phase and phase derivative measurements near the singularity for highly sensitive refractive index detection. The theoretical predictions performed with the Thin Film Theory are in agreement with the experimental findings from the conducted measurements.

Within the theoretical-experimental collaboration between the Nanoplasmonics Group (Facultad de Ciencias, UNAM) and the Biophotonics Group (Instituto Nacional de Astronomía, Óptica y Electrónica), I am working on the calculations and writing process of two manuscripts for submission as peer-reviewed publications:

- **Title: Theoretical and numerical approach to substrate effects in random plasmonic metasurfaces**
  Authors: Jonathan A. Urrutia-Anguiano, Isabel Y. Rojas-Martinez, Juan P. Cuanalo-Fernández, Alejandro Reyes-Coronado, Rubén Ramos-García, and Svetlana Mansurova

Status: Review between collaborators

Description: When analyzing the optical response of metasurfaces, discrepancies between experimental measurements and theoretical predictions are often qualitatively attributed to the influence of the substrate. In this study, we compare experimentally obtained reflectivity spectra of disordered metasurfaces composed of spherical gold nanoparticles (AuNPs) with predictions from the Coherent Scattering Model (CSM). Deviations from the experimental results are further examined through Finite Element Method (FEM) simulations of a single AuNP. By quantitatively estimating the contribution of multiple scattering and of image multipoles induced in the substrate, our findings, supported by experimental data, FEM simulations, and CSM predictions suggest that the gold nanoparticles are partially embedded in the substrate. This partial embedding effectively alters the apparent size of the nanoparticles and increases the surface coverage of the metasurface.

- **Title: Spectral shift in the strong couple regime: Red- and blueshift in plasmonic disordered metasurfaces**
  Authors: Juan P. Cuanalo-Fernández, **Jonathan A. Urrutia-Anguiano**, Irving Gazga-Gurrión, Nikolai Korneev, Alejandro Reyes-Coronado, Rubén Ramos-García, and Svetlana Mansurova
  Status: Writting process
  Description: The optical response of metasurfaces composed of disordered gold nanoparticles can be modeled using multiple scattering theories, such as the Coherent Scattering Model and the Thin Film Theory. Under Attenuated Total Reflection conditions, the resonance of these metasurfaces exhibited a spectral shift in response to changes in the refractive index of the surrounding medium of the nanoparticles. In addition to the well-known redshift, both theoretical predictions and experimental observations have also revealed a blueshift. This blueshift is believed to be an effect of strong coupling, as it occurs only when the angle of incidence is near the critical angle. Additionally, from a theoretical framework, it is studied if the blueshift is modeled only by multiple scattering theories, or if effective medium theories can reproduce such spectral behavior.

Chapter 5

Project Status and Next Stages

Research Progress

In summary, the progress achieved during the first year of my PhD studies includes the development of numerical code of analytical expressions for modeling disordered metasurfaces based on Multiple Scattering Theories, as well as the implementation of a parameter fitting method and the use of numerical simulation in specialized software. In addition, I have obtained experimental data from a first set of optical measurements of metasurfaces to further validate my theoretical findings. During this period, a peer-reviewed article on the Topological Darkness phenomenon observed in disordered metasurfaces was submitted and published. A manuscript based on the results presented in Section 2.2 is currently under internal revision between collaborators before submitting.

Proposed Workplan

This Section outlines the planned timeline for my research project focusing on expanding the results reported in Section 2.2 regarding partial embedding of metasurfaces and the spectral blueshift discussed in Section 3.2, from both theoretical and experimental standpoints. Additionally, I am planning a short fellowship to perform a sensitivity test at INAOE for the proposed disordered metasurface designs for biosensing applications. The continuation of my current investigations into substrate effects in disordered nanoparticle ensembles aims to contribute further to the understanding of optical interactions at the nanoscale under realistic scenarios.

- **January – June 2025**:
  - **Theory:** Expand the Thin Film Theory for ensembles of spherical particles with a non-neglectable size distribution based on the methodologies reported in Ref. [24] and Ref. [45].
  - **Experiment:** Perform first phase change of light reflected by a disordered plasmonic metasurface with elliposometric techniques to samples with a completed morphological characterization.

- **Writing:** Submit to publication the manuscript on substrate effects on disordered metasurfaces.

- **July – December 2025**:
  - **Theory:** *i)* Research on radiative correction within the dipolar approximation [46, 47] to explain the observed blueshift. *ii)* Investigate homogenization techniques and perform numerical simulations of partial embedding gold nanoparticles under oblique illumination. By matching these results, include them into the employed Multiple Scattering Theories via the refractive index surrounding spherical particles.
  - **Experiment:** Synthesize and morphologically characterize partially embedded metasurfaces with controlled degrees of incrustation.
  - **Writing:** *i)* Continue with the writing process of the manuscript on the spectral blueshift. *ii)* Begin the writing of a manuscript on partially embedded metasurfaces.

- **January – June 2026**:
  - **Theory:** Explore the inclusion of the substrate effects in the Coherent Scattering Model through the image method as done for one particle in Ref. [27].
  - **Experiment:** Based on theoretical and fabrication results, begin calibration for a phase-based biosensor with a disordered metasurface of spherical gold nanoparticles.
  - **Writing:** Begin the writing process of my PhD dissertation based on metasurfaces under realistic conditions.

- **July 2026 – December 2027**:
  - Pursue further research on the optical interaction of nanoparticles in disordered arrays, including non-linear effects [48] and radiative cooling [49].
  - Undergo the processes and fulfill the requirements established by the Posgrado en Ciencias Físicas to obtain the PhD degree.

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